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BOOK OF ABSTRACT

DIGITAL TRANSFORMATION, GREEN ENERGY, AND ADVANCED MATERIALS FOR A SUSTAINABLE SOCIETY

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"Digital Transformation, Green Energy, and Advanced Materials for a Sustainable Society"

Editors:
Zuli Rodhiyah
Fitra Wahyuni, M.Si

Reviewers:

- Dr. rer. nat Lucky Zaehir Maulana, S.Si., M.Sc.
- Ari Andrianti, S.Kom. M.Kom.

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SYNTHESIS AND CHARACTERIZATION OF MG-AL-NI LAYERED DOUBLE OXIDE (LDO) CATALYST

Nazarudin (1,2) , Intan Lestari (3), Hari Wiki Utama (4), Ulyarti (2,4), Sal Prima Yudha S (5), Vicka Andini (5), Fathan Bahfie (6)

1 Chemical Engineering Department, Faculty of Sciences and Engineering, University of Jambi,Indonesia
2 Centre of Excellence on Bio-Geo Material and Energy, University of Jambi, Indonesia
3Chemistry Department, University of Jambi, Indonesia
4Geological Engineering, Jambi University, Jambi, Indonesia
4 Acricultural Technology Department, Faculty of Agriculture, University of Jambi,Indonesia
5Chemistry Department, University of Bengkulu, Indonesia
6 Research Center of Mineral Technology, National Research and Innovation Agency of Indonesia, Lampung, Indonesia

This study presents the synthesis and characterization of Mg–Al–Ni Layered Double Hydroxide (LDH) and its derived Layered Double Oxide (LDO) catalysts prepared with varying nickel loadings (0.25–1.5 wt%) through a co-precipitation and intercalation approach. The catalysts were characterized using X-ray diffraction (XRD), X-ray fluorescence (XRF), and scanning electron microscopy (SEM) to investigate the structural, compositional, and morphological features. XRD analysis confirmed the successful formation of a hydrotalcite-like structure, with basal reflections corresponding to (003), (006), (012), (015), (018), and (110) planes, indicating well-ordered layered crystallinity. Incorporation of Ni^{2+} into the Mg–Al lattice up to approximately 1 wt% preserved high crystallinity and homogeneous substitution, whereas excessive loading ($\geqslant$1.25 wt%) led to peak broadening and reduced intensity, signifying partial loss of structural order and possible NiO segregation. Elemental analysis revealed a steady increase in Ni content proportional to the nominal loading, supporting effective metal incorporation within the LDH framework. SEM images showed that moderate Ni addition enhanced surface roughness and porosity, while excessive Ni promoted particle agglomeration. These results demonstrate that controlled nickel doping optimizes both the structural integrity and active-site distribution of Mg–Al–Ni LDO catalysts. The resulting material shows strong potential for application in catalytic cracking of used cooking oil and other biomass-derived feedstocks for biofuel production due to its synergistic acid–base and redox properties.

Keywords: Catalyst, Layered Double Hydroxide, Mg-Al LDH, Mg-Al-Ni LDH, Biofuel

INFLUENCE OF PLASTICIZER TYPE ON STARCH-BASED NANOCOMPOSITE FILMS REINFORCED WITH CMC AND ZNO

Rara Ayu Lestary, Ra'ida Raudhatussya'rifah, Rosmawati Sipayung, Sri Mundarti, Defia Indah Permatasari, Chindy Mauliza Duana

Universitas Jambi

The development of eco-friendly composite films from renewable resources represents a crucial strategy to reduce dependence on petroleum-based plastics. In this study, carboxymethyl cellulose (CMC) derived from oil palm empty fruit bunches (OPEFB) was utilized along with starch and zinc oxide nanoparticles (ZnO NPs) to fabricate biodegradable composite films. Two types of plasticizers, glycerol and sorbitol, were incorporated at varying concentrations of 20%, 30%, and 40% to evaluate their influence on the physical properties of the films, including density, thickness, moisture content, water absorption, and tensile strength. The results revealed that both plasticizers significantly affected the structural and physical characteristics of the films. Glycerol addition decreased density from 1.413 g/cm^3 to 0.463 g/cm^3 and increased moisture content and water absorption, whereas sorbitol addition increased density from 1.330 g/cm^3 to 1.559 g/cm^3 and reduced moisture content and water absorption. The formulation containing 20% glycerol showed optimal results for higher density and water resistance, while 40% sorbitol provided superior performance with enhanced mechanical integrity and lower water sensitivity. Overall, the type and concentration of plasticizer play a critical role in determining the physical properties of OPEFB-derived CMC/starch/ZnO nanocomposite films, demonstrating their potential as environmentally friendly packaging materials to replace conventional plastics.

carboxymethyl cellulose; oil palm empty fruit bunches; sorbitol; glycerol; composite film

A NOVEL SYNTHESIS ROUTE AND FUNCTIONAL GROUP ANALYSIS BY FTIR SPECTROSCOPY OF GRAPHENE OXIDE FROM INDIGENOUS COAL AND GRAPHITE

Lucky Zaehir Maulana, Valda Ayu Nanda, Rista Mutia Anggraini, Muhammad Ficky Afrianto, Fiqri Al Faruqi, Febri Berthalita Pujaningsih, Alrizal, Willy Bima Alfajri dan Frastica Deswardani

Universitas Jambi

This study reports a novel, cost-effective synthesis route for preparing Graphene Oxide (GO) using readily available indigenous coal from Jambi and natural graphite as dual carbon precursors. Graphene Oxide is a crucial intermediate material for producing graphene-based composites, and utilizing local, low-cost raw materials offers significant economic and scalability advantages over relying solely on commercial-grade graphite. The synthesis was successfully performed via a modified Hummers' method optimized for the unique chemical structure of the indigenous coal. The primary objective of this research was to confirm the successful oxidation and characterize the resulting oxygen-containing groups in the synthesized GO sheets. Fourier Transform Infrared (FTIR) Spectroscopy was employed as the key characterization technique. The FTIR analysis of the synthesized GO from both coal and graphite exhibited characteristic stretching vibrations, confirming the extensive oxidation of the carbon lattice. The FTIR results demonstrate that the novel synthesis method effectively introduces a high density of functional groups onto the carbon surface, which is essential for GO's applications in composites and chemical sensing. This work confirms the potential of local coal as a viable, alternative feedstock for high-quality GO production.

Graphene, nanomaterial, FTIR

FOURIER TRANSFORM INFRARED (FTIR) SPECTROSCOPIC ANALYSIS OF HYDROXYAPATITE SCAFFOLDS OBTAINED FROM CHANNA STRIATA AND SCOMBEROMORUS COMMERSONI FISH BONES

Rista Mutia Anggraini
Ria Anjelina
Febri Berthalita Pujaningsih
Muhammad Ficky Afrianto
Samsidar
Ichy Lucya Resta
Lucky Zaehir Maulana
Frastica Deswardani

Department of Physics, Universitas Jambi, Jambi 36361, Indonesia

Hydroxyapatite (HAp) has been successfully obtained from Channa striata and Scomberomorus commersoni bones. HAp scaffolds are potential materials for bone tissue engineering due to their adjustable characteristics suitable for specific biomedical applications. The freeze-drying method was employed in scaffold fabrication because it minimizes contamination and prevents carbon formation that may result from combustion, as the process operates at very low temperatures. A natural polymer derived from honeycomb was used to enhance the mechanical properties of the scaffold. Variations in the addition of honeycomb biopolymer were applied during the process. One of the characterizations performed was FTIR analysis, which was conducted to identify the functional groups formed. Variations in honeycomb concentration were found to influence the increase in FTIR absorption intensity. The functional groups identified include PO_4^{3-} and CO_3^{2-}.

scaffold, hydroxyapatite, fish bone, freeze-drying, honeycomb

ANALYZING THE FACTORS INFLUENCING GENERATION Z STUDENTS' ENTREPRENEURIAL INTEREST AT JAMBI UNIVERSITY USING SEM-PLS

Sarmada 1, Gusmanely.Z 2, Yuliana Safitri 3, Cut Multahadah 4, Nabilah Shahadah Putri*

Universitas Jambi

Unemployment is one of the major issues faced by Indonesia and must be addressed immediately, as it directly affects public welfare and economic growth. This issue also exists in Jambi Province, which faces similar challenges in creating adequate employment opportunities for its population. One of the solutions to reduce unemployment is by promoting entrepreneurship among Generation Z—encouraging them to be creative, innovative, and capable of utilizing digital technology to create new business opportunities. This study aims to analyze the factors influencing the entrepreneurial interest of Jambi University students using the Structural Equation Modeling–Partial Least Squares (SEM-PLS) approach. A quantitative survey method was employed, with data collected through a Likert-scale questionnaire. The exogenous variables tested include social and family environment, innovation and creativity, and technology, with entrepreneurial interest as the endogenous variable. The analysis results indicate that all exogenous variables have a significant effect on entrepreneurial interest, with an R^2 value of 0.59, meaning that 59% of the variation in entrepreneurial interest can be explained by the model. Among these, the factor of innovation and creativity shows the most dominant influence, with a parameter value of 0.352. These findings highlight the importance of strengthening innovation and creativity in fostering students' entrepreneurial interest.

SEM-PLS, SEM-PLS; Entrepreneurial Interest; Social and Family Environment; Innovation and Creativity; Technology.

MULTI-MODEL HYBRID FRAMEWORK FOR AUTOMATED MALARIA DETECTION AND SPECIES CLASSIFICATION

Akhiyar Waladi, Nindy Raisa Hanum, Yogi Perdana, Hasanatul Iftitah, Fitra Wahyuni, Rahmad Ashar

Universitas Jambi

Malaria remains a critical global health challenge with over 200 million annual cases, yet traditional microscopic diagnosis is time-consuming and requires expert pathologists. This study proposes a multi-model hybrid framework combining YOLO (v10-v12) for detection and six CNN architectures (DenseNet121, EfficientNet-B0/B1/B2, ResNet50/101) for classification, validated on two public MP-IDB datasets comprising 418 images across 8 classes (4 Plasmodium species and 4 lifecycle stages). The proposed shared classification architecture generates ground truth crops once and reuses them across all detection methods, enabling efficient model training. YOLOv11 achieves 93.10% mAP@50 with 92.26% recall on species detection and 92.90% mAP@50 with 90.37% recall on lifecycle stages. For classification, EfficientNet-B1 reaches 98.80% accuracy (93.18% balanced accuracy) on species despite 45:1 class imbalance, while EfficientNet-B0 achieves 94.31% accuracy (69.21% balanced accuracy) on lifecycle stages with 54:1 imbalance. A notable finding is that smaller EfficientNet models (5.3-7.8M parameters) outperform larger ResNet variants (25.6-44.5M parameters), with EfficientNet-B1 achieving 10.5 percentage points higher balanced accuracy than ResNet101 (93.18% vs 82.73%) despite 5.7× fewer parameters, challenging the "deeper is better" paradigm. The system addresses extreme class imbalance through Focal Loss (α=0.25, γ=2.0), achieving minority class F1-scores of 76.9% on P. ovale with perfect recall and 92.3% on schizont, though trophozoite (51.6%) and gametocyte (57.1%) remain challenging. Implemented on consumer-grade GPU hardware, the framework demonstrates practical feasibility for deployment in resource-constrained settings, though clinical use requires validation on larger datasets and field-collected samples

Malaria detection, Deep learning, YOLO, Class imbalance, Medical imaging, Plasmodium species, Lifecycle stages

HETEROGENEOUS VISION-LANGUAGE ENSEMBLE FOR MULTI-ACCOUNT INSTAGRAM ENGAGEMENT PREDICTION

Jefri Marzal
Muhammad Razi
Miranty Yudistira
Akhiyar Waladi
Hamzah Alghifari

Universitas Jambi

Predicting social media engagement remains challenging due to heterogeneous user behaviors and complex multimodal signals. Existing approaches predominantly employ single convolutional neural network (ConvNet) architectures with simple text embeddings, overlooking the semantic understanding capabilities of modern vision-language transformers and the benefits of architectural diversity. We propose a heterogeneous ensemble framework combining three ConvNet architectures (ResNet-50, ConvNeXt, EfficientNet-B0) with two vision-language transformers (CLIP, Vision Transformer) for Instagram engagement prediction. Through systematic ablation studies across 22 experiments on 10,520 posts from 8 Instagram accounts, we demonstrate that architectural heterogeneity yields superior performance compared to homogeneous ensembles. Our analysis reveals that account-specific features provide the largest single improvement (10.2% MAE reduction), while vision-language models contribute complementary semantic information (+0.42% improvement over ConvNet-only ensemble). The framework achieves mean absolute error of 170.77 likes with Ridge meta-learning employing negative ensemble weights for systematic error correction—an advantage unavailable to constrained methods. Notably, we observe 30-fold performance variance across accounts (MAE: 10.18–295.29), indicating highly heterogeneous engagement dynamics. Uncertainty analysis demonstrates well-calibrated ensemble disagreement (r=0.51 with prediction error), enabling confidence-based deployment where low-uncertainty predictions achieve MAE of 19.33. Systematic failure mode analysis identifies posts without faces as exhibiting 46% higher error, providing actionable insights for future feature engineering. Our results establish state-of-the-art performance while maintaining training efficiency at 4.9 minutes on consumer-grade GPUs.

Vision-language models, multimodal fusion, ensemble learning, Instagram engagement prediction, social media analytics, CLIP, vision transformer, heterogeneous architectures, meta-learning, deep learning

AUTOMATION OF EMPLOYEE ATTENDANCE SYSTEM USING CONVOLUTIONAL NEURAL NETWORK (CNN) AND YOLO ALGORITHM FOR FACE RECOGNITION WITH DEEP LEARNING APPROACH

Pradita Eko Prasetyo Utomo, Edi Saputra, Mohammad Ilhami, Benedika Ferdian Hutabarat, Ulfa Khaira

Universitas Jambi

Deep learning is widely used in computer vision, a branch of machine learning that aims to teach systems to understand and interpret images and videos. Object detection is a computer-assisted task related to detecting examples of visual objects of a specific class. The goal of object detection is to develop computational models and techniques. Deep learning algorithms are widely used for various innovations in computer vision, one of which is the automated attendance system. This study aims to create an innovation in the form of an automated attendance system, namely an automatic attendance recording system that utilizes CCTV to capture images/videos of employees entering the office and performs facial detection and recognition processes by applying deep learning algorithms and then records the presence of recognized employees automatically. In this study, OpenCV is used as a facial detection algorithm from input images/video streams that will produce input images for the facial recognition process. Facial recognition and classification are made using the YOLO and CNN algorithms. The training process for the formation of object recognition models will utilize a dataset in the form of a collection of facial images/photos that have been collected so far through a web-based facial attendance application.

cnn;yolo;face recognition,attendance

ADAPTIVE PRACTICAL SCHEDULING SYSTEM BASED ON GENETIC ALGORITHM WITH MULTI-OBJECTIVE OPTIMIZATION APPROACH

Pradita Eko Prasetyo Utomo, Benedika Ferdian Hutabarat, Niken Rarasati

Universitas Jambi

Practical scheduling is a crucial challenge in higher education, particularly in the Faculty of Science and Technology (FST) of Jambi University, which has numerous study programs, laboratory-based courses, and limited resources. The manual or semi-digital scheduling process often results in scheduling conflicts, inefficient space utilization, and inconsistencies with lecturer and student preferences. This research develops SmartLab Scheduler, an adaptive practical scheduling system based on a genetic algorithm with a multi-objective optimization approach. This system is designed to generate optimal schedules by considering various objectives simultaneously, such as minimizing scheduling conflicts, optimizing space utilization, and accommodating instructor preferences. The genetic algorithm is implemented with a chromosome representation containing important parameters such as days, hours, laboratory space, courses, and faculty. A fitness function is designed to evaluate solutions based on the level of violation of mandatory constraints (hard constraints) and optional preferences (soft constraints). This system is integrated into a user-friendly web/desktop-based information platform, so it can be used directly by academic administrators. This research produces a reliable, adaptive, and replicable system on a wider scale. The main outputs are a ready-to-use SmartLab Scheduler system, scientific publications in accredited journals (SINTA 2), conference/seminar proceedings with ISBNs, and intellectual property rights.

scheduling, genetic, practical, optimation

DESIGN AND IMPLEMENTATION OF AN IOT-BASED REAL-TIME MONITORING SYSTEM FOR OPTIMIZING THE GROWTH OF CHILI PLANTS (CAPSICUM ANNUUM L.)

Andre Rabiula, Edi Saputra, Pradita Eko Prasetyo Utomo, Benedika Ferdian Hutabarat, Mahya Ihsan

Universitas Jambi

Chili (Capsicum annuum L.) is one of the most important horticultural commodities in the world; however, its production is often constrained by unpredictable environmental conditions and suboptimal management practices. Precision agriculture based on the Internet of Things (IoT) offers a potential solution to address these challenges. This study aims to design, develop, and test a real-time monitoring system for essential growth parameters of chili plants. The system integrates a series of sensors to measure soil moisture, air temperature, air humidity, and light intensity. Data collected by the microcontroller are processed and wirelessly transmitted to a cloud platform for storage and analysis. An interactive web dashboard was developed to visualize both historical and real-time data, enabling users to perform remote monitoring. The testing results indicate that the system is capable of acquiring and transmitting data accurately and reliably with minimal latency. This system holds great potential to assist farmers in making data-driven decisions, optimizing the use of resources such as water and fertilizers, and ultimately improving the yield and quality of chili production.

Chili, Internet of Things (IoT), Precision Agriculture, Monitoring System, Smart Farming

PHYTOCHEMICAL SCREENING AND ANTIOXIDANT ACTIVITIES OF ETHANOLIC EXTRACTS OF METAXYA PARKERI (HOOK. & GREV.) J.SM.

Uni Baroroh Husnudin[1], Dina Erliana[2], Munifilia Ekasari[2], Andini Vermita Bestari[1], and Anggari Linda Destiana[1]

[1] Biology, Faculty of Science and Technology, Universitas Jambi, Jambi, Indonesia
[2] Chemistry, Faculty of Science and Technology, Universitas Jambi, Jambi, Indonesia
Corresponding author email: uni.baroroh.h@unja.ac.id

Metaxya parkeri (Hook. & Grev.) J.Sm. is a fern species whose pharmacological potential has not been extensively explored. This study aimed to screen for phytochemical compounds and evaluate the antioxidant activity of ethanolic extracts from the petiole and pinnae of M. parkeri. Extraction was performed using the maceration method with 96% ethanol. Qualitative phytochemical screening was conducted to detect alkaloids, flavonoids, saponins, tannins, and terpenoids. Antioxidant activity was determined using the DPPH (2,2-diphenyl-1-picrylhydrazyl) method, expressed as the Inhibitory Concentration 50 (IC50). Phytochemical screening revealed that both petiole and pinnae extracts tested positive for alkaloids, tannins, saponins, and flavonoids. The ethanolic extracts exhibited weak antioxidant activity, with IC50 values of 160.54 ± 4.63 ppm for the petiole and 117.81 ± 0.61 ppm for the pinnae, whereas the positive control (ascorbic acid) demonstrated strong activity (44.75 ± 0.67 ppm). These results indicate that M. parkeri contains bioactive compounds. However, their concentration in the crude ethanolic extract may be insufficient to produce strong antioxidant activity, suggesting the need for further extraction or purification methods.

Metaxya parkeri; antioxidant; DPPH; phytochemicals; ethanolic extract

PHYTOCHEMICAL PROFILE AND ANTIOXIDANT ACTIVITY OF DAEMONOROPS DRACO (WILLD.) BLUME RESIN EXTRACT FROM JAMBI, INDONESIA

Fitra Wahyuni1, Ade Adriadi2, Anggari Linda Destiana3, Ashif Irvan Yusuf4, Revis Asra5

Department of Biology, Faculty of Science and Technology, Universitas Jambi

This study aims to characterize the phytochemical constituents and evaluate the antioxidant activity of Daemonorops draco (Willd.) Blume resin, locally known as jernang, a traditional medicine widely used for wound healing in Sumatra, Indonesia. Resin samples were macerated using 96% ethanol and screened qualitatively for major secondary metabolites. Phytochemical profiling revealed the presence of flavonoids, phenolics, and triterpenoids, while alkaloids and saponins were absent. Antioxidant capacity was determined through the DPPH radical scavenging assay. The extract exhibited a concentration-dependent inhibition of DPPH radicals (R^2 = 0.96) with an estimated IC_{50} of 160.4 µg/mL, indicating moderate antioxidant potential. This activity correlates with the abundance of flavonoid compounds, known to neutralize reactive oxygen species and support tissue repair. The findings scientifically confirm the ethnopharmacological use of jernang resin as a natural antioxidant source. These data provide a foundation for subsequent studies involving FRAP and HRBC assays, as well as in vivo wound-healing models to explore its anti-inflammatory and regenerative mechanisms.

Daemonorops draco; jernang; phytochemical screening; antioxidant activity; DPPH assay

DEVELOPMENT OF KOMBUCANG (SECANG WOOD KOMBUCHA) AS A FUNCTIONAL HERBAL DRINK: ANTIOXIDANT AND ANTIHYPERURICEMIC ACTIVITIES

Ahmad Sazali, Ashif Irvan Yusuf, Hasna Ul Maritsa, Fitra Wahyuni

Universitas Jambi

Kombucha is a fermented beverage known for its various biological activities, including antioxidant and uric acid-lowering effects. Caesalpinia sappan L. (sappan wood) contains active compounds such as brazilin and flavonoids, which have potential as natural antihyperuricemic agents. This study aimed to evaluate the antihyperuricemic activity of Kombucang (Kombucha of Sappan Wood) on mice (Mus musculus) induced with potassium oxonate, assess the in vitro antioxidant activity of Kombucang using the DPPH method, and examine its feasibility as a functional herbal beverage from pharmacological and organoleptic aspects. The research was conducted using a completely randomized design (CRD) with six treatment groups: a normal control without treatment (K), a positive control induced with potassium oxonate (KP), a negative control treated with potassium oxonate and allopurinol (KN), and three Kombucang-treated groups with different doses: 0.26 ml/20 g body weight (P1), 0.52 ml/20 g body weight (P2), and 0.78 ml/20 g body weight (P3). Each treatment consisted of five replications and was administered for 14 days. The results showed that administration of Kombucang at a dose of 0.52 ml/20 g body weight (P2) significantly reduced uric acid levels in mice induced with potassium oxonate compared to the positive control group. The DPPH antioxidant assay demonstrated that Kombucang possessed strong antioxidant activity, supporting its potential as a functional herbal beverage with beneficial pharmacological effects and favorable organoleptic properties.

Kombucha; Kombucang; Sappanwood; Antihyperglycemic; Caesalpinia sappan

PHENOTYPIC SPECTRUM AND DISORDER CANDIDATE ANALYSIS IN ECTRODACTYLY CASES IN SOUTH SULAWESI, INDONESIA

Firdaus

Faculty of Biology UGM, Yogyakarta, Indonesia

Background: Ectrodactyly is a rare congenital disorder characterized by reduced digits and a deep median cleft in the hands and/or feet. This condition has wide genetic heterogeneity and phenotypic expressivity, and can be either isolated or syndromic. There is no literature that reveals the phenotypic spectrum of ectrodactyly from South Sulawesi. Objective: To characterize the phenotypic spectrum and identify candidate disorders in ectrodactyly cases in South Sulawesi, Indonesia. Methods: Four individuals with ectrodactyly from non-consanguineous families were characterized phenotypically and radiologically. Phenotypic examination included head & neck, chest, skeletal, skin, nails & hair, and supported by plain radiography to assess the bone structure of the hands and feet. Each phenotypic character was annotated using The Human Phenotype Ontology (HPO) and then analyzed with Phenotype Similarity Tools using the Ancestor Information Content (AIC) metric to identify candidate disorders. Results: All individuals showed bilateral involvement with a broad phenotypic spectrum, including tridactyly to hexadactyly with median cleft, syndactyly, camptodactyly, hallux valgus, brachydactyly, hallux duplication, nail dysplasia, phalangeal aplasia, triphalangeal thumb, and metatarsal and metacarpal synostosis. No anomalies were found outside the hands and feet. Candidate disorder analysis identified Split Hand-Foot Malformation 4 (SHFM4; MIM 605289) as the strongest candidate. Conclusion: This study demonstrates a highly variable phenotypic spectrum of ectrodactyly in South Sulawesi, with SHFM4 as a candidate disorder. Molecular confirmation through TP63 gene sequencing is recommended to uncover the underlying genetic mechanism.

Ectrodactyly, South Sulawesi, Phenotypic Spectrum, Phenotype Similarity Tools, SHFM4

A REVIEW OF MICROBIAL DEGRADATION OF PLASTIC: ENZYMES AND ORGANISMS

Zuli Rodhiyah and Silvia Annur Annisa

UNIVERSITAS JAMBI

This review synthesizes research on systematic analysis of microbial applications in degrading plastic and microplastic waste, focusing on bacteria, fungi, actinomycetes, and enzymes such as PETase, cutinase, and laccase to address persistent plastic pollution and remediation challenges. The review aimed to evaluate microbial taxa involved in biodegradation, benchmark enzymatic activities, identify consortia strategies, elucidate metabolic pathways, and assess environmental influences on degradation efficiency. Literature was selected based on comprehensive inclusion of experimental, metagenomic, enzymatic, and biotechnological studies spanning global environmental microbiology and synthetic biology, analyzed through comparative and critical frameworks. Findings reveal diverse microbial communities, including key bacterial and fungal genera, with consortia exhibiting synergistic enzymatic cooperation that enhances polymer breakdown. Enzymatic mechanisms of PETase, cutinase, and laccase demonstrate variable catalytic efficiencies, improved by protein engineering and immobilization, though challenges remain for non-PET plastics. Metabolic pathways for monomer assimilation and gene regulation are partially elucidated, with environmental factors such as temperature and pH critically modulating microbial and enzymatic performance. Integration of multi-omics and synthetic biology advances supports enhanced biodegradation but highlights scalability and methodological standardization gaps. These findings collectively underscore the potential of microbial and enzymatic systems for sustainable plastic waste remediation while emphasizing the need for integrative approaches to optimize efficacy and application in diverse environmental contexts.

Plastic Degradation; Microorganisms; Bioremediation; Environmental Microbiology.

IN VITRO COMPATIBILITY EVALUATION OF TRICHODERMA HARZIANUM, BACILLUS SUBTILIS, AND PSEUDOMONAS FLUORESCENS ON KING'S B AND PDA-NA MEDIA

Putri Dwi Mulyani, Triyuni Verza Abdilla, Hasna Ul Maritsa

Universitas Jambi

Trichoderma harzianum, Bacillus subtilis, and Pseudomonas fluorescens are widely known as effective biological control agents that can be developed as environmentally friendly alternatives for plant disease management. However, the combined application of these microorganisms requires compatibility assessment to prevent antagonistic interactions within a consortium. This study aimed to evaluate the compatibility of T. harzianum with B. subtilis and P. fluorescens on King's B and PDA–NA media using a dual culture assay. Observations included the presence of inhibition zones as indicators of antibiosis and the diameter of T. harzianum colonies measured at 24 and 48 hours to determine the percentage inhibition (PI). The results showed that on King's B medium, clear inhibition zones were observed in both combinations, with PI values of 57.04% for T. harzianum × B. subtilis and 62.89% for T. harzianum × P. fluorescens at 48 hours, indicating strong antagonism. In contrast, no inhibition zones were detected on PDA–NA medium, and the PI values were considerably lower (10.96% and 18.40%, respectively), suggesting a more compatible interaction. These findings indicate that the three microorganisms may be applied together as a consortium under environmental conditions that do not promote excessive antagonistic metabolite production, as represented by PDA–NA medium or similar field conditions. This study provides fundamental insight into the development of multi-agent biocontrol formulations.

Trichoderma harzianum; Bacillus subtilis; Pseudomonas fluorescens; compatibility; biocontrol.

DEVELOPMENT OF A PROTOTYPE HERBAL FISH MEDICINE FROM SAPPAN WOOD EXTRACT (CAESALPINIA SAPPAN L.): AN ANTIMICROBIAL INNOVATION TO REDUCE MORTALITY RATES IN AQUACULTURE

Ahmad Sazali, Ade Adriadi, Hasna Ul Maritsa, Ashif Irvan Yusuf

Biology Studi Program, Jambi University

Kegagalan panen dalam budidaya ikan, khususnya di Provinsi Jambi, sering disebabkan oleh infeksi bakteri patogen seperti Aeromonas hydrophila, A. Sobria, E. ictaluri, dan Streptococcus iniae. Penggunaan antibiotik sintetis secara berlebihan telah menimbulkan resistensi mikroba serta berdampak negatif terhadap lingkungan dan kesehatan manusia. Penelitian ini bertujuan untuk mengembangkan prototipe obat herbal ikan berbasis ekstrak kayu secang (Caesalpinia sappan L.) dalam bentuk larutan perendaman yang memiliki aktivitas antibakteri tinggi terhadap bakteri penyebab penyakit ikan. Metode penelitian meliputi ekstraksi kayu secang dengan pelarut etanol 96%, formulasi tiga jenis larutan perendaman menggunakan kombinasi bahan tambahan (Tween 80, PEG 400, dan gliserin), uji aktivitas antibakteri secara in vitro terhadap empat bakteri patogen ikan. Luaran penelitian yang ditargetkan meliputi satu prototipe obat herbal ikan siap pakai, publikasi pada jurnal nasional terakreditasi Sinta 2, diseminasi kepada pembudidaya ikan, serta peningkatan Tingkat Kesiapterapan Teknologi (TKT) dari level 4 menjadi level 6. Hasil penelitian menunjukkan bahwa Formula 3 (mengandung gliserin sebagai bahan tambahan) menghasilkan zona hambat terbesar terhadap keempat bakteri uji, menunjukkan aktivitas antibakteri paling tinggi dan stabilitas larutan yang baik. Formula 3 memiliki zona hambat rata-rata sebesar 17 cm untuk keempat bakteri Aeromonas hydrophila, A. Sobria, E. ictaluri, dan Streptococcus iniae dengan katergori kuat. Dengan demikian, Formula 3 ditetapkan sebagai formulasi terbaik untuk dikembangkan menjadi prototipe obat herbal ikan yang efektif dan aman digunakan dalam sistem budidaya perikanan.

Obat Ikan, Caesalpinia sappan, Antimikroba, Patogen Ikan, Budidaya Perikanan

MORPHOLOGICAL AND PHYSIOLOGICAL RESPONSES OF COFFEA LIBERICA TO NPK AND SWIFTLET MANURE ON ULTISOL

Elis Kartika1,3, Lizawati1,3, Adzuan Handi Duansyah1, Yusnaidar2,3, Madyawati Latief2,3, Samsidar3,4, Indra Lasmana Tarigan2,3*

1Department of Agroecotechnology, Faculty of Agriculture, Universitas Jambi, Indonesia
2Department of Chemistry, Faculty of Science and Technology, Universitas Jambi, Indonesia
3The University Centre of Excellences, E2-KOLIM, Universitas Jambi
4Department of Physic, Faculty of Science and Technology, Universitas Jambi, Indonesia

Ultisol soils are characterized by low fertility, acidic pH, and limited nutrient availability, which constrain the early growth of Coffea liberica seedlings. This study aimed to evaluate the effects of combining NPK fertilizer and swiftlet manure on the morphological and physiological growth of C. liberica seedlings grown in Ultisol soil. The experiment employed a completely randomized design (CRD) with a single factor consisting of six treatments: (1) control (no fertilizer), (2) 100% recommended NPK dose, and combinations of 50% NPK with swiftlet manure at (3) 150 g, (4) 300 g, (5) 450 g, and (6) 600 g per polybag. Observed parameters included plant height, stem diameter, number of leaves, root and shoot dry weight, total leaf area, chlorophyll content, and N, P, K uptake. Results showed that the combined application of NPK fertilizer and swiftlet manure significantly affected both morphological and physiological growth of C. liberica seedlings. The optimal response was obtained with 300 g of swiftlet manure combined with 50% NPK, producing the highest plant vigor, nutrient uptake efficiency, and biomass accumulation while improving Ultisol soil properties. This combination effectively reduced NPK use by half without compromising growth performance, suggesting a sustainable, cost-efficient, and environmentally friendly fertilization strategy for C. liberica seedling production on acidic soils

Coffea liberica; Ultisol; NPK fertilizer; swiftlet manure; seedling growth; nutrient uptake; sustainable fertilization

DIVERSITY AND BIOPROSPECTING POTENTIAL OF LIGNICOLOUS FUNGI IN THE JAMBI UNIVERSITY EDUCATIONAL FOREST

Anggari Linda Destiana, Jessica Maranatha Tambun, Elvira Aulia Daryanti, Putri Dwi Mulyani, Hasna Ul Maritsa, Anggit Prima Nugraha, and Tia Wulandari

University of Jambi

Wild lignicolous fungi are a group of fungi that play a crucial role in wood decomposition and nutrient cycling in forest ecosystems. Furthermore, these lignicolous fungi represent a hidden gem for herbal medicine. This research aims to investigate the diversity and bioprospecting potential of wild lignicolous fungi located at the periphery of the University of Jambi Educational Forest. The study employed a combined exploratory-analytical method and literature review, with stages including fungal sampling and collection, morphological characterization and identification, and a literature-based study on their bioprospecting potential. The results revealed the collection of 17 wood fungus samples with distinct morphological characteristics. Ten of these samples were successfully identified, while seven remain unidentified. All identified samples belong to the Phylum Basidiomycota, Order Agaricomycetes, and are divided into 3 Classes (Agaricales, Polyporales, Auriculariales) and 6 Families (Mycenaceae, Polyporaceae, Fomitopsidaceae, Auriculariaceae, Schizophyllaceae, and Agaricaceae). The fungi from these six families have diverse applications, including use as food sources, herbal medicine ingredients, and decomposers. In conclusion, the wood fungi obtained from the edge of the University of Jambi Educational Forest are diverse and possess significant bioprospecting value worthy of further development.

Basidiomycota, Bioprospecting, Fungi, Lignicolous, University of Jambi

ZOOPLANKTON COMMUNITY STRUCTURE AND ITS RELATIONSHIP TO PHYSICO-CHEMICAL PARAMETERS IN THE SWAMP ECOSYSTEM OF RAWA BENTO, KERINCI

Olifia Rifani, Tedjo Sukmono, Dawam Suprayogi

Biology Study Program, Faculty of Science and Technology, Universitas Jambi, Jambi, Indonesia
Doctoral Student of Biology, Faculty of Biology, Gadjah Mada University, Yogyakarta, Indonesia

This study aims to analyze the diversity, abundance, and community structure of zooplankton and its relationship with the physico-chemical parameters of the waters in Rawa Bento, Kerinci Regency, Jambi Province. The Bento Swamp Ecosystem is a highland swamp that has an important ecological function as a buffer for water systems, carbon storage, and habitat for various aquatic biota. The research was carried out in February-May 2023 using a quantitative descriptive approach through field survey methods at three stations representing differences in anthropogenic activity levels. Zooplankton samples were taken with a plankton net measuring 64 µm and identified using a binocular microscope. The physico-chemical parameters measured include temperature, pH, brightness, flow velocity, dissolved oxygen (DO), biochemical oxygen requirement (BOD_5), and total suspended solids (TSS). The identification results found 9 species of zooplankton classified into four classes and three phyla (Rotifera, Protozoa, and Ciliophora), with the dominance of Rotifera (Eurotatoria and Monogononta) contributing about 80% to the total community. The diversity index (H') value ranged from 2.058–2.093 (medium category), the equality index (E) was high (0.936–0.953), and the dominance (C) was low (0.132–0.140), indicating a stable community. Correlation analysis showed that pH and TSS were positively correlated very strongly with zooplankton abundance, while temperature and BOD_5 were negatively correlated very strongly. These results indicate that the water conditions of Rawa Bento still support zooplankton life, despite light anthropogenic pressures. These findings are expected to be basic data for ecological monitoring and sustainable management of highland swamp conservation.

Water Quality; Rawa Bento; Community Structure; Zooplankton

CONSERVATION STATUS AND HABITAT ZONATION OF FRESHWATER FISH IN MERANGIN GEOPARK JAMBI INDONESIA

Tedjo Sukmono1,Pradita Eko Prasetyo Utomo, Yoppie Wulanda, Lauura Hermala Yunita, Febrina Rolin, Rizky Janatul Magwa, Yusyam Leni, Musadat.

1Biology Study Program, Faculty of Science and Technology, University of Jambi, l. Jambi Muaro Bulian Km.15 - Mendalo Darat, Jambi, Indonesia
2Information Systems Study Program, Faculty of Science and Technology, University of Jambi, l. Jambi Muaro Bulian Km.15 - Mendalo Darat, Jambi, Indonesia
3Fishery Product Technology Study Program, Faculty of Animal Husbandry, University of Jambi, l. Jambi Muaro Bulian Km.15 - Mendalo Darat, Jambi, Indonesia
4Fishery Resource Utilization Study Program, Faculty of Animal Husbandry, University of Jambi, l. Jambi Muaro Bulian Km.15 - Mendalo Darat, Jambi, Indonesia
5PT REKI Hutan Harapan, Indonesia

The Merangin Geopark, located in Merangin Regency, Jambi Province, is now part of the Global Geopark Network (GGN) under the auspices of UNESCO (Mei 24, 2023) and plays an important role in preserving biodiversity, particularly within aquatic ecosystems. One of its key components, the Batang Merangin River, serves as a natural habitat for various freshwater fish species and contributes significantly to maintaining ecological balance. However, anthropogenic activities, especially illegal gold mining (locally known as PETI), have altered the aquatic environment, posing serious threats to the sustainability of endemic species. This study aimed to identify the conservation status and map the habitat zonation of conservation-valuable fish species in the Batang Merangin River within the Merangin Geopark area. Data collection was conducted in September 2025 at four sampling stations (Air Batu Village, Fosil Kayu, Muara Karing, and Teluk Wong) using a field survey approach. The results identified 26 fish species, including two categorized as Vulnerable (Bagarius yareli and Lobocheilos falcifer) and four as Data Deficient (Glyptothorax formelicus, Mystus sabanus, Osteochilus festivus, and Tor tambra) based on the IUCN Red List. Spatial mapping revealed that the Fosil Kayu and Muara Karing stations are priority conservation zones, while Teluk Wong functions as a natural spawning habitat. In conclusion, the Batang Merangin River hosts several fish species at high risk of extinction according to the IUCN status, emphasizing the need for targeted conservation efforts. The spatial analysis highlights critical zones that should be prioritized for ecosystem-based conservation management to sustain fish biodiversity within the Merangin Geopark.

Bagarius yareli; Conservation; IUCN; Lobocheilos falcifer; The Merangin Geopark;

CYPRINIDAE DIVERSITY AND ECOSYSTEM SERVICES IN ACEH: A SYSTEMATIC STUDY FOR SUSTAINABLE FISHERIES MANAGEMENT

Andri Yusman Persada, Andhika Puspito Nugroho, and Rury Eprilurahman

Faculty of Biology, Universitas Gadjah Mada, Yogyakarta, Indonesia

Family Cyprinidae is the largest group of freshwater fish in the world. It plays an important role in maintaining the balance of ecosystems, including nutrient cycles, food webs, and community stability. This study aims to identify the biodiversity, distribution, and conservation status of Cyprinidae in Aceh, as well as their role in supporting ecosystems, communities, and social sustainability. Data were compiled from Global Biodiversity Information Facility (GBIF), Encyclopedia of Life (EOL), FishBase, IUCN Red List, and related scientific articles up to September 2025. A descriptive analysis of geographic distribution, species richness, and conservation status was conducted based on IUCN Criteria. The results show that there are 41 species from 12 Cyprinidae genera in Aceh. These species are dominated by Rasbora, Osteochilus, Tor, and Barbonymus. The spatial distribution shows that the districts and cities in South Aceh and Middle Aceh have the highest biodiversity. Most species are in the Least Concern category, but some genera, such as Barbodes and Tor, are highly threatened due to habitat degradation and human activity. The high diversity of the Cyprinidae family demonstrates its important role in providing ecosystem services, such as a food source, regulating ecosystems, and providing cultural value. This study emphasizes the need for an ecosystem-based approach to fisheries management that integrates biodiversity conservation with socio-economic sustainability. Integrating local knowledge and modern science could produce a Cyprinidae-based bioeconomic model that supports both conservation and community welfare in the inland waters of Aceh

Aceh, biodiversity, conservation, cyprinidae, sustainable fisheries

IDENTIFICATION OF AMPHIBIANS OF THE ORDER ANURA IN MUHAMMAD SABKI URBAN FOREST PARK, JAMBI CITY

Anggit Prima Nugraha (1), Tia Wulandari (1), Dawam Suprayogi (1), Andini Vermita Bestari (1), Rizky Desryan Syah (2), Verataty Gultom (2), Louisa Bella Br Tampubolon (2), Despita (2)

(1) Biology Study Program, Department of Mathematics and Natural Sciences, Faculty of Science and Technology, Universitas Jambi, Jambi, Indonesia
(2) Student of the Biology Study Program, Department of Mathematics and Natural Sciences, Faculty of Science and Technology, Universitas Jambi, Jambi, Indonesia

Green Open Spaces (GOS) play an important role in maintaining urban ecosystems and provide ecological, social, and economic benefits for local communities. The Muhammad Sabki Urban Forest Park (Taman Hutan Kota Muhammad Sabki) in Jambi City is one of the GOS areas that still contains a natural forest ecosystem. The order Anura is one of the components of this ecosystem and also functions as an environmental bioindicator. This study aimed to characterize the morphology and assess the species diversity of the order Anura in Muhammad Sabki Urban ForestPark. The survey method used was Visual Encounter Survey (VES), applied to all habitat types in each zone. Data collection was conducted at 07.00-10.00 and 19.00-22.00 Western Indonesian Time. Observations were carried out in high-use, moderate-use, and low-use zones. Photograph of anuran individuals were taken, and species were identified based on morphological character using field guides. The results consist of morphological character description and conservation status. Based on the observations, 40 individuals representing 10 species from five families were recorded. The five families encountered in this survey were Bufonidae, Dicroglossidae, Ranidae, Microhylidae, and Rhacophoridae. The ten species recorded were Hylarana erythraea, Hylarana baramica, Hylarana glandulosa, Limnonectes microdiscus, Limnonectes blythii, Duttaphrynus melanostictus, Polypedates leucomystax, Kurixalus chaseni, Microhyla sp. 1, and Microhyla sp. 2.

Anura; Bioindicator; Muhammad Sabki Urban Forest Park.

ARTHROPODA COMMUNITY STRUCTURE IN SWEET CORN-TURMERIC INTERCROPPING AGROECOSYSTEM

Wilyus Wilyus1,, Dwi Ristyadi1, Asniwita1, Fuad Nurdiansyah1*

Universitas Jambi

Arthropod diversity plays an important role in maintaining the balance and ecological resilience of agroecosystems. Intercropping systems provide more diverse habitats and resources, which can attract various arthropod species and influence community composition. This study aimed to investigate the diversity of arthropods in the sweet corn–turmeric intercropping agroecosystem. The research was conducted at the Experimental Farm of the Faculty of Agriculture, University of Jambi, and arthropod identification was carried out in the Plant Pest Laboratory. The observation variables focused on species diversity, abundance, and ecological functional groups of arthropods. Arthropod samples were collected at the early of the sweet corn vegetative phase using pitfall traps, yellow sticky traps, and sweep nets. Data were analyzed using the Shannon–Wiener diversity index (H′), dominance index (D), Evenness index (E), and true alpha diversity (^1D) = $e^{H'}$). The results showed that the values of H′, D, E, and ^1D were 2.64, 0.09, 0.88, and 13,96, respectively. The ecological functional structure of arthropods consisted of herbivores (25.0%), predators (61.1%), parasitoids (2.8%), pollinators (5.6%), and decomposers (2.8%). These results indicate that the sweet corn–turmeric intercropping agroecosystem supports high arthropod diversity, a stable community structure, and provides favorable conditions for the conservation of natural enemies, particularly predators.

Arthropod diversity; intercropping; sweet corn; turmeric; ecological roles

CLIMATE CHANGE AND THE FUTURE OF COASTAL FISHERIES: ECOLOGICAL SHIFTS, GEOPOLITICAL IMPLICATIONS, AND PATHWAYS TO SUSTAINABLE DEVELOPMENT

Muhammad Irsyad Abiyusfi Ghafari1,2, Namira Risqi Putri Muquita3*, Djumanto4, Eko Agus Suyono1, and Andhika Puspito Nugroho1*

1 Faculty of Biology, Gadjah Mada University, Indonesia
2 Life as Naturalist, Indonesia
3 Department of International Relations, Faculty of Social Science and Political Science, Gadjah Mada University, Indonesia
4 Department of Fisheries, Faculty of Agriculture, Gadjah Mada University, Indonesia

Climate change is reshaping the world's coastal and marine ecosystems, posing complex challenges for ecological stability, food security, and maritime governance. This review synthesizes current knowledge on the ecological, social, and geopolitical dimensions of climate change impacts on coastal fisheries. Ecologically, ocean warming, sea-level rise, and habitat degradation—particularly in mangrove and coral reef systems—are driving shifts in species distribution and altering ecosystem productivity. These biophysical changes cascade into socio-economic and geopolitical tensions as fish stocks migrate across national boundaries, challenging existing governance regimes and creating new arenas for negotiation and conflict. Drawing upon interdisciplinary literature from marine ecology, political geography, and sustainable development studies, this paper critically examines how ecological transformations intersect with economic dependency and regional power dynamics in coastal states. It further identifies gaps in current adaptation frameworks, highlighting the need for integrated, science-based, and inclusive policy responses that link local resilience with global cooperation. The review concludes by proposing pathways toward sustainable coastal fisheries grounded in ecosystem-based management, maritime diplomacy, and equitable blue economy strategies. By bridging ecological evidence and governance innovation, this study contributes a holistic perspective on how coastal fisheries can adapt to a rapidly changing climate while maintaining ecological integrity and social justice.

climate change; ecological shifts; blue economy; maritime governance; sustainable development

ANALYSIS OF MICROPLASTIC CONTENT IN AMBIENT AIR IN THE CAMPUS AREA (CASE STUDY: MENDALO CAMPUS, JAMBI UNIVERSITY)

Zahra Nurulita Amanda1, Febri Juita Anggraini2, Fernando Mersa Putra3

Universitas Jambi

Microplastic pollution has become a global environmental issue, including microplastic contamination in ambient air, where air plays an important role as an indicator of environmental quality. This study aims to analyze the abundance and characteristics of microplastics contained in the ambient air of the Mendalo Campus, Jambi University. The sampling method uses a gravimetric technique with a High Volume Air Sampler (HVAS) at three locations representing various activities within the campus area, namely: Campus Forest, Campus Main Gate, and Campus Waste Disposal Site. The results of the abundance calculation for one of the samples showed a value of 228,709 particles/gram and 91.304 particles/gram, which indicates significant microplastic contamination in the campus environment, with the types of microplastics found being fibers, fragments, and films. These findings confirm that the campus area, although partially covered by good vegetation, is not completely free from the presence of microplastics in the air, therefore requiring serious attention in plastic waste management and air pollution mitigation.

Microplastics; Ambient Air; Abundance; Environmental Pollution.

ESTIMATION MODEL OF MANGROVE FOREST BIOMASS USING SATELLITE IMAGERIES IN JAMBI PROVINCE, INDONESIA

Eva Achmad, Hamzah, Fazriyas

Department of Forestry, Jambi University, Faperta Building, Jambi University, Jambi, Indonesia

Mangrove forests are having quite high biomass content. Currently, data regarding biomass content stored in mangrove forests, especially in Jambi Province, is still limited. Carbon calculations using field surveys are a very good method, but calculations using this method are considered less efficient in terms of time and cost, especially if the area to be studied has a large area. An alternative method for calculating biomass can be done by utilizing remote sensing technology, namely with Landsat 8 satellite imagery. Estimating biomass stores using satellite imagery can be done using a vegetation index approach and model making, where in this research the vegetation index used is NDVI and the model chosen is the model quadratic with the model equation, namely Y = -1315.754 + 9452.41 NDVI2 with an R2 value of 65.5%. Model selection is based on the highest score value which considers SA, SR, RMSE, Bias, R2, and s. The mangrove forest biomass stock map is divided into several classes, namely biomass storage <1 ton/ha, then classes 1-100 (ton/ha), 100-200 (ton/ha), 200-300 (ton/ha), 300- 400 (ton/ha), >400 (ton/ha), where the class with a value range of >400 (ton/ha) dominates, reaching 3015, 98 ha or 30.04% of the mangrove area in Jambi Province.

estimation model; forest biomass; Landsat imagery; mangrove forest; NDVI

HUMIC ACID COATED BIOCHAR IMPROVES RETENTION IN AND REDUCES NITROGEN EMISSION FROM FERTILIZED SOILS OF PALM OIL PLANTATIONS

Damris Muhammad
Edwin Permana
Diah Riski Gusti
Intan Lestari
Martina Asti Rahayu

Universitas Jambi

Biochar, a carbon-rich and porous material produced from organic waste biomass, has become a new trend in overcoming various environmental isues. However, direct virgin biochar application has limited capacity. This research aims to synthesis humic acid (HA) coated biochar and to evaluate HA-coated biochar in retaining N in and reducing N emission from soil of palm oil plantations. HA coated-biochar was obtained by autoclaving HA solutions (0.25% and 1.0%) for 30 minutes and then biochar was added with stirring. The mixture was then dropped into 1% and 5% CaCl2 solutions for cross-linking formation. The HA-coated biochar (0.00, 0.75, 1.50, and 3.00% (w/w) were mixed with 500g soil and incubation was carried out for 60 days. Factions of soil-HA-coated biochar and gas from the bootle head space were collected 4 times at the day 0, 20, 40, and 60. The solid was treated with water and 0.1M HCl at a ratio of 1:5 for the determination of N retention and the gas was analyzed with GC for the determination of N2O gas emission. HA-coated biochar amendment significantly increased N retention ($p<0.05$) and significantly reduced N2O emission from the soil ($p<0.05$). The increase in retention and the decrease in emission of N from the soils simultantly reduced N loss more than 40% compared to the soil without HA-coated biochar amendment.

humic acid; biochar; nitrogen; soil; incubation

IMPLEMENTATION OF A WEB AND MOBILE-BASED DIGITAL ATTENDANCE SYSTEM TO IMPROVE EMPLOYEE MANAGEMENT EFFECTIVENESS AT PRAKTEK DOKTER PATTIMURA, JAMBI CITY

Muhammad Razi A., Tri Suratno, Benedika Ferdian Hutabarat, Dedy Setiawan, Dewi Lestari, Yolla Noverina, Willy Bima Alfajri

Universitas Jambi

The main problem faced by Praktek Dokter Pattimura in Jambi City is the employee attendance system, which still relies on manual recording and standalone fingerprint machines that are not connected to the internet. This condition causes delays in attendance recording, data inaccuracy, and a high risk of manipulation. This Community Service Program (PPM) aims to assist the partner institution in developing and implementing a digital attendance system based on web and mobile technology, integrated with face recognition and Global Positioning System (GPS). The implementation method uses a Prototype model consisting of several stages: system design, application development, testing, implementation, and user assistance. The results show that the digital attendance system improves administrative efficiency, data accuracy, and employee discipline. User satisfaction reached 88.09% for the mobile application and 85.82% for the web application, indicating that the system was well accepted by users. Therefore, this digital attendance system is proven to be effective and can serve as a model for adopting digital technology in healthcare service management.

Digital Attendance, Face Recognition, GPS, Clinic, Human Resource Management

INTRODUCTION OF MINING VIRTUAL REALITY (MINING VR) TO GEOLOGY AND MINING STUDENTS OF VOCATIONAL SCHOOL MUHAMMADIYAH JAMBI CITY

*Muhammad Ikrar Lagowa *1), Aditya Denny Prabawa 1), Yudi Arista Yulanda 1), Jarot Wiratama 1), Bagus Adhitya 2), Yosa Megasukma 1), Dimas Agung Permadi 1), Affif Farras Chan1), Siti Sadariah 1),*

Universitas Jambi

The employment of vocational high school graduates in the mining sector remains relatively low compared to other educational levels. With the continued expansion of the mining industry in Indonesia, particularly in Jambi Province, the establishment of vocational high schools with Mining Geology majors in Jambi City is anticipated to contribute to the supply of workers. The scarcity of teachers, particularly those with a mining education background, can be a significant factor in the decline in the quantity and quality of these vocational high school graduates. This program seeks to introduce Mining Virtual Reality (Mining VR) as a contemporary learning methodology for mining. Utilizing this simulation program, students can engage directly with various mining activities within the classroom without incurring substantial financial and time expenditures. Furthermore, they gain access to restricted areas, such as underground mines and blasting zones. The intended outcomes of this program encompass: (1) augmenting student knowledge and comprehension of several mining learning modules, (2) fostering students' practical skills in the utilization of digital technology through Virtual Reality Simulators, and (3) disseminating activities in the form of articles on social media platforms and comprehensive reports. This program is expected to enhance the quality of graduates of Muhammadiyah Vocational High Schools specializing in Mining Geology.

mining virtual reality, learning digitalisation, Vocational High Schools majoring in Mining Geology

A WEB-BASED INFORMATION SYSTEM FOR DIGITALIZING ADMINISTRATIVE PERMIT SERVICES AT UNIVERSITY Y

Mochammad Arief Hermawan Sutoyo, Renaldi Yulvianda, and Willy Bima Alfajri

Universitas Jambi

The Indonesian government's budget efficiency policy has compelled University Y to reduce its operational expenditures. This financial constraint has created an urgent need to modernize the university's conventional administrative processes, particularly the permit management procedures, which remain dependent on paper-based documentation, limited human resource capacity, and manual archiving systems that are prone to errors. To address these challenges, digitalizing the process must be done, this study proposes the design of an integrated web-based permit information system. The system development adopts the prototyping methodology, and its effectiveness is evaluated using the System Usability Scale (SUS) to ensure the system operates properly and enhances user convenience. The evaluation results indicate a score of 83, suggesting that the prototype meets user expectations and is ready for implementation.

Software Enginering, HCI, Administrative System, Digitalization

ANALYSIS OF SHALLOW GROUNDWATER QUALITY AS AN INDICATOR OF SEAWATER INTRUSION IN THE NORTHERN COAST OF JAVA

Merisa R. Mawaddah, Risnaliyah Nuriil Tadersi, Febiyora Chandra Kirana, Andicho Haryus Wirasapta

Universitas Jambi

This study investigates the quality of shallow groundwater as an indicator of seawater intrusion in Yosorejo Village, Batang Regency, located on the northern coast of Java. This study evaluates physicochemical parameters for Total Dissolved Solids (TDS), Electrical Conductivity (EC), and pH, incorporating geological factors to investigate preliminary indicators of saline intrusion. Groundwater sampling and analysis took place in September-October 2023 using seven domestic wells distributed at various distances from the coastline (0-4 km) and topographic elevations (0-10 meters above sea level). GIS mapping of TDS and Electrical Conductivity revealed peaks for TDS with a maximum reading of 1400 ppm and Electrical Conductivity with a maximum reading of 1970 µS/cm, indicating that saline intrusion is well under way. It also confirmed increasing tendencies towards decreasing elevations nearshore. Groundwater pH varies between 7 to 8.4, within tolerable standards but slightly basic inland along hydrolysis and lithological reactions. Geologically, this site is primarily covered by very porous layers of alluvial deposit silt sand, representing aquifers amenable to saline waters. This work presents its originality by doing integrated spatial hydrogeological analyses to identify saline intrusion affected areas within 0-500 meters onshore. This work contributes to the development of environmental physics and hydrogeological research by confirming empirically for the imbalanced interface between seawater and groundwaters on tropical low-lying areas, pressing for alleviating strategies to secure groundwaters' sustainability.

Groundwater Quality; Coastal Sediment; Seawater Intrussion

IDENTIFICATION OF SUBSURFACE STRUCTURES USING THE WENNER-SCHLUMBERGER CONFIGURATION GEOELECTRIC METHOD AT THE PEMATANG JERING TEMPLE SITE IN MUARO JAMBI

Husnul Hamdi, Indah Khairun Nisa, Maison

Universitas Jambi

A study entitled "Identification Of Subsurface Structures Using The Wenner-Schlumberger Configuration Geoelectric Method At The Pematang Jering Temple Site In Muaro Jambi" has been conducted. Indonesia possesses a rich cultural heritage characterized by the diversity of its ethnic groups, cultures, and historical relics scattered across various regions, including Jambi Province. One of the most significant historical sites in this area is the Muaro Jambi Temple Complex, an archaeological site estimated to contain at least 83 temple structures, although only 9 have been successfully restored to date. The existence of this site highlights its considerable archaeological potential, emphasizing the need for further investigation to uncover the buried structures and artifacts. This study aims to model subsurface structures using two-dimensional resistivity data and to identify the locations of temple structures composed of buried brick formations in the Pematang Jering Temple Site, within the Muaro Jambi area. The method employed is electrical resistivity using the Wenner-Schlumberger configuration. Measurements were conducted along five survey lines, and 2D inversion of the data using RES2DINV software revealed resistivity values ranging from 0.00885 to 82.3 Ωm. Indications of brick structures were found along all survey lines, with resistivity values ranging from 17 to 82.3 Ωm. These findings suggest the presence of potential hidden archaeological structures at the research site.

Resistivity, Geoelectricity, Wenner-Schlumberger, Subsurface

SUPERGENE ALTERATION AND COPPER MINERALIZATION ANOMALY IN PERALUMINOUS (S-TYPE) GRANITOIDS AT TELUK TILAN SITE, MERANGIN GEOPARK, JAMBI

Yudi Arista Yulanda, Jarot Wiratama, Muhammad El Hakim, and Muhammad Faisal Seprizal

Universitas Jambi

This study aims to identify geochemical and mineralogical anomalies within the Tantan Granite Formation at the Teluk Tilan site (Sample TL) in Merangin Geopark, Jambi, to understand the potential for metal enrichment. Petrographic, mineragraphic, and X-Ray Fluorescence (XRF) analyses were conducted on a granitoid sample from this specific location. Geochemically, Sample TL is classified as a Peraluminous (S-type) Granite, characterized by a molar A/CNK value of 1.282. This finding is considered anomalous because the rock exhibits the presence of Copper (Cu) mineralization with a concentration 0.1%. Mineragraphic analysis confirmed an association of metallic minerals, including chalcopyrite (CuFeS2), pyrite (FeS2), and most significantly, the presence of the secondary minerals Covellite (CuS) and Goethite (FeO(OH)). The occurrence of covellite indicates that this granitoid has undergone a post-magmatic supergene alteration process. This supergene process (weathering) triggers the leaching of primary copper and the re-precipitation of secondary copper, potentially leading to higher grades. The results support the interpretation that the Tantan Granite Formation records a complex magmatic evolution and holds indicators of a prospective base metal mineralization system within the Merangin Geopark area.

Tantan Granite; Copper; Supergene Alteration; Covellite; Merangin Geopark

HISTOPATHOLOGY OF GILLS AND MUSCLES IN SEVERAL FISH SPECIES CONTAMINATED BY GOLD MINING WASTE IN LEBONG REGENCY

Dian Fita Lestari, Vestidhia Yunisya Atmaja, Santi Nurul Kamilah

Department of Biology, Faculty of Mathematics and Natural Sciences, University of Bengkulu, Bengkulu, Indonesia

Lebong Regency is one of the gold mining area in Bengkulu province. Heavy metals are used in gold mining processes to separate gold from rock and sand. The waste is subsequently discharged into the river, affecting the water bodies near the mine in Lebong Regency. Heavy metals can cause organ damage in a variety of ways in fish. Increasing heavy metal contamination can result in morphological, physiological, and biochemical alterations in fish. Heavy metal pollution can cause structural and functional damage to many fish organs. Alterations in histopathological structures can inform adaptive responses to changing environments or in cases of severely altered conditions. Histopathological examination is one of the examination techniques by studying abnormal changes in cells or tissues which is used to determine the diagnosis of the effects of environmental contaminants on fish. The aim of this research was to examine histopathological of gills and muscles in several types in fish impacted heavy metal from gold mining waste. The result showed that the effects of heavy metals on fish can be seen from the photomicrograph of the histopathological structure of the gills and muscles that showing histological damage to the gills including fusion secondary lamellae, epithelial lifting, hyperplasia and hypertrophy. Additionally, the gills of T.tricopterus are infected with parasites. Meanwhile, photomicrograph of the muscles histology show the muscle atrophy, edema, necrosis, and fatty degeneration are common symptoms of damage fish muscles. Heavy metal waste is one of the pollutant that can causes alteration in fish organs.

alteration, fish, gold mining, heavy metals

ETHNOBOTANY OF MEDICINAL PLANTS AMONG THE ANAK DALAM TRIBE IN NYOGAN VILLAGE, JAMBI, INDONESIA: CULTURAL KNOWLEDGE AND POTENTIAL FOR HERBAL-MEDICINE DEVELOPMENT

Revis Asra; Ade Adriadi; Fitra Wahyuni; Yogi Irawan Pandiangan; Febriyanti Munthe; Defrizal; Izu Andry Fijridiyanto

Universitas Jambi; Bappeda Provinsi Jambi; Pusat Riset Biosistematika Dan Evolusi - BRIN

This collaborative ethnobotanical study documents and analyzes the traditional knowledge of medicinal plant utilization among the Anak Dalam (SAD) community permanently settled in Nebang Parah and Nyogan Hamlets, Nyogan Village, Muaro Jambi Regency, Indonesia. Fieldwork employed a descriptive qualitative approach through direct observation, in-depth interviews with 54 informants (including 1 tribal healer), participatory observation, and herbarium collection for species identification. Twenty-one plant species belonging to 18 families were recorded, predominantly from wild habitats. Frequently used taxa included Annona muricata, Cratoxylum arborescens, Melastoma malabathricum, Curcuma longa, and Tinospora crispa, whose active compounds (flavonoids, xanthones, curcuminoids) demonstrate proven anti-inflammatory, antipyretic, and antioxidant properties. Quantitative indices: Cultural Significance Index (ICS) and Plant Part Value (PPV), revealed that leaves constituted the most-used organ (48%), while species with high ICS values reflected strong ritual and healing importance. The findings show that SAD's indigenous pharmacopoeia represents an invaluable biocultural asset currently endangered by rapid socio-cultural change. This research provides a scientific foundation for the preservation of traditional medicinal knowledge and identifies several high-value species for further phytochemical and pharmacological exploration toward standardized Jambi-based herbal products. The study contributes to sustainable utilization of biodiversity and strengthens community-based health resilience through local herbal innovation.

ethnobotany; medicinal plants; Anak Dalam tribe; traditional knowledge; herbal medicine

TECHNIQUE FOR PRODUCING LOW CHOLESTEROL SPICED MAYONNAISE WITH ADDITION OF MONO-DIASIL GLYCEROL (MDAG) COEMULSIFIER

Mursalin Mursalin, Fitry Tafzi, Tasya Dwi Safitri

Department of Agricultural Technology, Faculty of Agriculture, Jambi University, Jalan Raya Jambi-Muara Bulian Km.15 Mendalo Darat, Jambi, Indonesia

Synthesis of cocoa butter equivalent (CBE) from a mixture of RBDPO olein fraction (refined olein, ROL) and soft-PMF olein fraction (mid olein, MOL) at 5 weight ratios (70/30); (60/40); (50/50); (40/60); (30/70), catalyzed by 0.2% w/w sodium methoxide, have been studied. Characterization was carried out before and after interesterification, including triacylglycerol composition (AOCS Method Ce 5b-89, 2003), Solid Fat Content (AOCS Method Cd 16b - 93:2017), and Slip Melting Point (AOCS Method Cc 3-25, 2005). Chemical interesterification caused some TAGs were increased, some were decreased, and several new TAGs were formed. Synthesis of target TAGs (POS, SOS) increased as the proportion of MOL in the substrate were increased. ROL/MOL mixture at a weight ratio of 50/50; 40/60; and 30/70 have characteristics similar to cocoabutter (CB) and could be considered as CBE. CBE with TAG markers of more than 70% and CBE index of more than 60% produced from a mixture formulation ROL/MOL at a ratio of 30/70. This finding confirmed that CBE could be produced from mid olein (a by-product of hard PMF production) and RBD olein. Based on this finding, the economic value of ROL and MOL in the future may be increased and their usage map may also be changed.

low cholesterol mayonnaise; MDAG; palm-based product; seasoning

CHEMICAL INTERESTERIFICATION OF MIXTURE OF RBDPO AND SOFT-PMF OLEIN FRACTIONS FOR SYNTHESIZING COCOA BUTTER EQUIVALENTS

Mursalin Mursalin, Fitry Tafzi, Yernisa Yernisa

Department of Agricultural Technology, Faculty of Agriculture, Jambi University

Synthesis of cocoa butter equivalent (CBE) from a mixture of RBDPO olein fraction (refined olein, ROL) and soft-PMF olein fraction (mid olein, MOL) at 5 weight ratios (70/30); (60/40); (50/50); (40/60); (30/70), catalyzed by 0.2% w/w sodium methoxide, have been studied. Characterization was carried out before and after interesterification, including triacylglycerol composition (AOCS Method Ce 5b-89, 2003), Solid Fat Content (AOCS Method Cd 16b - 93:2017), and Slip Melting Point (AOCS Method Cc 3-25, 2005). Chemical interesterification caused some TAGs were increased, some were decreased, and several new TAGs were formed. Synthesis of target TAGs (POS, SOS) increased as the proportion of MOL in the substrate were increased. ROL/MOL mixture at a weight ratio of 50/50; 40/60; and 30/70 have characteristics similar to cocoabutter (CB) and could be considered as CBE. CBE with TAG markers of more than 70% and CBE index of more than 60% produced from a mixture formulation ROL/MOL at a ratio of 30/70. This finding confirmed that CBE could be produced from mid olein (a by-product of hard PMF production) and RBD olein. Based on this finding, the economic value of ROL and MOL in the future may be increased and their usage map may also be changed.

chemically interesterification; cocoa butter equivalents; palm oi; triacylglycerol

ENHANCING THE CAPACITY OF MUSHROOM FARMERS THROUGH TRAINING ON THE PRODUCTION OF OYSTER MUSHROOM SOY SAUCE AND POWDER

Anggari Linda Destiana, Hasna Ul Maritsa, Putri Dwi Mulyani, Uni Baroroh Husnudin, Fitra Wahyuni, Anggit Prima Nugraha, Mahya Ihsan, and Ahmad Sazali

Biology Department, Science and Technology Faculty, Universitas Jambi

Oyster mushroom is an edible fungus widely cultivated in Jambi. Based on digital data searches, there are 9 MSMEs (Micro, Small, and Medium Enterprises) engaged in oyster mushroom cultivation in the region. One such MSME collaborating with the community service team from the Biology Study Program, Faculty of Science and Technology, Jambi University, is Ryan Jamur. Ryan Jamur MSME harvests approximately 30-50 kg of fresh oyster mushrooms per harvest cycle, which are then marketed fresh to several traditional markets in Jambi. Fresh oyster mushrooms (Pleurotus ostreatus) have a relatively short shelf life due to their high moisture content (80-90%) and rapid enzymatic activity. Therefore, this community service initiative aims to enhance the capacity of mushroom farmers through training on the production of soy sauce and powder from oyster mushrooms, which have a longer shelf life compared to fresh ones. This service activity consists of three main stages: (1) preparation (initial survey, activity planning, and coordination with partners), (2) implementation of training on the production of mushroom soy sauce and powder, and (3) activity evaluation. Based on the evaluation results, 72.73% of participants assessed that producing oyster mushroom powder and soy sauce is easy to do, indicating that the participants feel capable of producing them independently.

oyster mushroom, mushroom soy sauce, mushroom farmers, mushroom powder

CYSTEINE-CAPPED SILVER NANOPARTICLES AS A PROBE FOR COLORIMETRIC SENSING OF 3-MONOCHLOROPROPANE-1,2-DIOL

Martina Asti Rahayu, Arum Widia Putri, Dhian Eka Wijaya, Ayu Nadhila Safitri, M. Irhash Shalihin

Universitas Jambi

3-Monochloropropane-1,2-diol (3-MCPD) is a recognized food contaminant formed during the thermal processing of fats and oils, posing significant health concerns due to its potential toxicity. In this study, we present a rapid and sensitive colorimetric assay based on cysteine-modified silver nanoparticles (Cys-AgNPs) for the detection of 3-MCPD. The Cys-AgNPs were synthesized via chemical reduction and characterized by UV–Vis spectrophotometry. Upon interaction to 3-MCPD, the solution color changed visibly from yellow to pink within five minutes at 80 °C and pH 9.3, corresponding to a shift in the surface plasmon resonance (SPR) band. This phenomenon is attributed to the reaction between the amino group of cysteine and 3-MCPD, forming N-(2,3-dihydroxypropyl)-amino acid moieties grafted onto the nanoparticle surface (3-MCPD–Cys–AgNPs). Quantitative analysis was achieved using UV–Vis measurements, yielding a linear response over the concentration range of 1–5 μg mL^{-1}, with a calculated detection limit of 0.126 μg mL^{-1}. The pronounced color transition enables unassisted visual detection of 3-MCPD even at 1 μg mL^{-1}. Overall, this Cys–AgNP colorimetric probe provides a simple, rapid, and cost-effective approach for monitoring 3-MCPD in food samples, offering potential for on-site quality and safety assessment.

3-MCPD, silver nanoparticles, colorimetric detection, food contaminant, rapid assay.

TERMITICIDAL ACTIVITY OF METHANOL EXTRACT OF TETRACERA INDICA LEAVES AND DETERMINATION OF CHEMICAL COMPONENTS USING GC-MS

Dina Erliana1, Munifilia Ekasari2, Uni Baroroh3, Nurul Pratiwi4, Melyanda5, Sindi Haryani6, Tio Ersa Relyansyah7, Muhammad Irfan Nur widad8

Universitas Jambi

The purposes of this research to investigate the termiticide activity of methanol extract of Tetracera indica against Coptotermes curvignathus Holmgren using the no choice test method, and to determine the chemical components of extract by using GC-MS. The stages of this research were: sampling, preparation extract of the Tetracera indica plant using the maceration method, phytochemical screening, determination of chemical components using GC-MS, and termiticidal activity testing. Termiticide activity tests showed that the percentage of termite mortality increased with the concentration applied. Concentrations of 20% and 25% resulted in the highest termite mortality, reaching 100%. Phytochemical screening results of flavonoids, saponins, tannins, steroids/terpenoids, and GC-MS analysis showed 26 chemical components, with the main compounds being methyl octadecanoate and methyl palmitate.

Tetracera indica, GC-MS, Termiticide

GREEN SYNTHESIS OF AG/TIO$_2$ COMPOSITE USING PINEAPPLE PEEL EXTRACT FOR DEGRADATION OF ORGANIC POLLUTANTS

Nurul Pratiwi, Muhammad Irhash Shalihin, Munifilia Ekasari and Fiqri Al Faruqi

Chemistry, Faculty of Science and Technology, Universitas Jambi, Jambi, Indonesia

This study aims to develop a green synthesis method for Ag/TiO_2 composites using pineapple peel waste as an eco-friendly bioreductor for organic pollutant treatment applications. The utilization of agricultural waste as a natural reducing agent is expected to replace hazardous chemicals in the synthesis process while supporting the principles of sustainable chemistry. The synthesis was carried out using a hydrothermal approach with variations in pineapple peel extract concentration to obtain composites with optimal morphology and photocatalytic activity. Characterization was performed using XRD, UV-Vis DRS, SEM-EDX, and FTIR to determine crystal structure, composition, and functional group changes. Photocatalytic activity was evaluated through the degradation of an organic pollutant model, methylene blue, under visible light irradiation. The results are expected to show that the presence of silver nanoparticles enhances light absorption and TiO_2 photocatalytic performance through the formation of surface plasmon resonance (SPR). This research is anticipated to produce an effective and environmentally friendly photocatalyst while adding value to pineapple peel waste as a source of functional biomaterials. Moreover, the outcomes are expected to serve as a foundation for further applied research in wastewater treatment and green material technology development.

Green synthesis; Ag/TiO_2 nanocomposite; Pineapple peel waste; Bioreduction; Photocatalytic degradation

CHELATING ZINC(II) IN CRUDE PHEOPHORBIDE GIVES IT CO_2-REDUCING CAPABILITY

Muhammad Irhash Shalihin
Panji Cakra Buana

Universitas Jambi

A pheophorbide-based photocatalyst prepared from water hyacinth (Pontederia crassipes) was developed for photocatalytic CO_2 reduction. Crude pheophorbide was prepared from the chlorophyll extract of fresh water hyacinth leaves by a sequential acid–base treatment. The successful conversion of chlorophyll to pheophorbide was indicated by the decreased absorbance intensity and slight red shift of the Qy band from around 662 nm to approximately 667 nm, indicating demetallation at the chlorophyll core as well as dephytylation. Preliminary photocatalytic investigation proved that the presence of a Zn center in pheophorbide resulted in its ability to convert CO_2 into CO and other organic compounds in water as the solvent under light irradiation. CO and organic molecules within a closed reaction system were detected and confirmed photocatalytic reduction of CO_2. However, bubbles formed during the process, probably owing to residual impurities carried in the crude chlorophyll precursor, degraded reactor stability. Additional purification and optimization therefore are required for further stability and effectiveness of the catalyst. Although this research is at the initial stage, this study demonstrates the feasibility of utilizing water hyacinth–derived pheophorbide–Zn complexes as potential organometallic photocatalysts for CO_2 reduction in support of green chemistry principles.

CO_2, Chlorophyll, Pheophorbide, Photocatalysis, Zinc

THE EFFECT OF PH AND BIOCHAR MASS VARIATIONS ON THE EFFICIENCY OF COD AND BOD ABSORPTION LEVELS IN WASTEWATER

Teguh Setyawan, Damris M, and Ummi Kalsum

Universitas Jambi

Biochar is a carbon-rich solid material produced by the conversion of organic waste through incomplete combustion or limited oxygen supply (pyrolysis). Wastewater is the waste water that is discharged from households, industry and other public places, and generally contains materials or substances that can be harmful to human health and disrupt the environment. This study aims to determine the greatest absorption efficiency of COD and BOD levels from wastewater that can be reduced through the adsorption process using biochar. The adsorption process was carried out using variations in weight of pH 2, 4, 6, 8, 10, and 0.5, 1, 1.5, 2, 2.5 grams to determine the optimum conditions for the greatest absorption efficiency. The analysis results showed the greatest absorption efficiency for COD levels at pH 6 of 96.953% and 1.5 grams of biochar of 96.33% and BOD levels at pH 8 of 95.589% and 2 grams of biochar of 95.41% by wastewater.

Biochar, Wastewater, COD, BOD

SYNTHESIS OF SO4-FE2O3-MGO CATALYST USING MICROWAVE IRRADIATION COMPARED TO CONVENTIONAL HEATING

Tamami Nukma Mujnadila, Lea Aprilyani, Rahma Amalia, Oki Alfernando, Ira Galih Prabasari, Rosmawati Sipayung, Sri Mundarti, Nita Widyastuti, Rizka Shafira

Jambi University

The demand for renewable energy has encouraged the development of heterogeneous catalysts to improve methyl ester production from waste cooking oil. Sulfated metal oxide catalysts exhibit strong acidity and good thermal stability, but their physical properties and catalytic activity are strongly affected by the activation method. SO4-Fe2O3-MgO catalysts were synthesized via co-precipitation with activation compared between microwave irradiation and conventional heating. Fe(NO3)3·9H2O and Mg(NO3)2·6H2O precursors were precipitated using a combination of NH4OH and NaOH as precipitating agents to reach pH 9–11, followed by 15 hours of aging, filtration, drying, and calcination at 600 °C. Sulfation was carried out by incipient wetness impregnation using 1 M (NH4)2SO4, then dried and calcined again at 500 °C. Catalyst characterization was limited to XRD and FTIR to determine crystal structure and functional groups. Catalytic activity was tested in the transesterification of waste cooking oil under optimized conditions of 65 °C for 150 minutes with a 1:12 methanol-to-oil ratio, using three systems: 1% homogeneous NaOH catalyst, 5% SO4-Fe2O3-MgO catalyst prepared by conventional heating, and 5% SO4-Fe2O3-MgO catalyst activated by microwave irradiation. Results indicated that microwave-activated catalysts exhibited higher crystallinity, more homogeneous sulfate distribution, and greater methyl ester yield compared to conventional heating and homogeneous NaOH base catalysts. These findings confirm that microwave irradiation is a more efficient method for enhancing the structural and acidic properties of SO4-Fe2O3-MgO catalysts and for improving catalytic performance in methyl ester production from waste cooking oil.

SO_4-Fe_2O_3-MgO, sulfation, microwave irradiation, methyl ester, waste cooking oil

STIRRING VARIATION EFFECT ON SILICA-CELLULOSE ADSORPTION EFFICIENCY FOR FFA REDUCTION

Intan Nandia Sakti1, Rosmawati Sipayung1, Oki Alfernando1, Ira Galih Prabasari1, Rahma Amalia1, Rara Ayu Lestary1, Nita Widyastuti1, Nazarudin1

Universitas Jambi

Used cooking oil contains high free fatty acid (FFA) levels that inhibit transesterification and promote saponification, making biodiesel purification difficult; therefore, an effective pretreatment is required to reduce FFA content to meet biodiesel feedstock standards. This study evaluates the adsorption performance of agricultural-waste-based bioadsorbents, namely rice husk ash as a silica source and corn cobs as a cellulose source, under batch conditions with 300 rpm stirring. Operational variations included adsorbent concentrations of 1–5%, temperatures of 30–60 °C, and contact times of 30–90 minutes. The adsorbents were produced through calcination of rice husk and alkaline treatment of corn cobs, followed by FTIR and XRD characterization to identify functional groups and structural phases. The results show that silica-rich rice husk ash exhibits superior performance, achieving an 81.44% FFA reduction at the optimum condition of 5% adsorbent concentration, 60 °C, and 30 minutes, resulting in a final FFA value of 0.20%. In contrast, cellulose-based corn cobs achieved only a 39.02% reduction at 5% concentration, 30 °C, and 90 minutes, with a final FFA level of 0.66%. FTIR confirmed active silanol and hydroxyl groups, while XRD revealed an amorphous silica structure with higher reactivity compared to the semi-crystalline cellulose of corn cobs. These chemical and structural differences explain why stirring more effectively enhances silica adsorption than cellulose. The novelty of this study lies in integrating stirring variation with comparative adsorbent behavior, contributing to applied physics through adsorption dynamics, surface interaction mechanisms, and material utilization. This research provides an environmentally friendly approach to waste valorization and supports physics education by demonstrating the relevance of material properties and mass-transfer phenomena in real-world energy applications

adsorbent; biodiesel; corncob; free fatty acid; rice husk

COMMUNITY ECO-ENZYME FERMENTATION FOR HOUSEHOLD ORGANIC WASTE

Ade Adriadi1, Revis Asra2, Mahya Ihsan3, Tia Wulandari4, Ahmad Sazali5 and Fitra Wahyuni6

Department of Biology, Faculty of Science and Technology, Universitas Jambi

This community engagement program addressed unmanaged household organic waste among Malay mothers in Ulu Gedong, Jambi, by transferring a low-cost, science-based fermentation technology to produce multi-purpose eco-enzyme. Objectives were to (i) improve knowledge and skills in source-level waste valorization, (ii) generate usable products (liquid eco-enzyme and compostable residues), and (iii) catalyze a resident-led recycling group. Methods combined participatory briefing, hands-on training, and a 30-day anaerobic fermentation using a 1:3:10 sugar:biowaste:water formulation, EM4 bio-activator, and a simple water seal air lock; monitoring was conducted through on site visits and messaging. Results included ~10 L matured eco-enzyme (golden-brown, pleasant acidic aroma) per batch, immediate household use as natural cleaner and plant tonic, and formation of the Ulu Gedong Recycling Mothers group. A post-activity survey (n=16) showed very high satisfaction and intent to continue practice (overwhelming "Strongly Agree/Agree"), indicating effective knowledge transfer and social acceptance. Novelty lies in a rapid, community owned, low resource anaerobic protocol that shortens maturation from ~90 to ~30 days while embedding behavior change and micro enterprise opportunities. Implications include measurable source reduction of organics, reduced chemical cleaner purchases, and a scalable green economy model for similar peri urban communities.

eco-enzyme; household organic waste; community engagement; anaerobic fermentation; green economy

SYNTHESIS AND APPLICATION OF AGRICULTURAL WASTE SILICA-CELLULOSE ADSORBENT FOR NON-AGITATED FFA ADSORPTION FROM WASTE COOKING OIL

Putri Ananda, Rosmawati Sipayung, Oki Alfernando, Ira Galih Prabasari, Nazarudin, Rahma Amalia, Rara Ayu Lestary, Nita Widyastuti

Universitas Jambi

Waste cooking oil is a promising and low-cost raw material for biodiesel production; however, its high Free Fatty Acid (FFA) content inhibits the transesterification process and lowers biodiesel yield. This study aims to develop a silica–cellulose composite bioadsorbent synthesized from rice husk ash and corn cob, and to evaluate its effectiveness in reducing FFA under non-agitated conditions. Silica was extracted through calcination and alkaline leaching, while cellulose was isolated via NaOH delignification. Both materials were crosslinked using glutaraldehyde and characterized using FTIR, and XRD to determine their functional groups, structural phases, and surface morphology. Characterization confirmed the integration of silica and cellulose through Si–O–Si, O–H, and C–O–C functional groups, an amorphous semi crystalline hybrid structure, and porous morphology favorable for adsorption. Adsorption experiments with varying dosage, temperature, contact time, and silica–cellulose ratios demonstrated that the highest efficiency occurred at 3–5% adsorbent dosage, 60–90 minutes contact time, and 45°C. Under optimal conditions, the bioadsorbent successfully reduced FFA levels to below 2%, meeting biodiesel pretreatment standards. The novelty of this research lies in the development of a low-cost, agricultural-waste-based composite functioning effectively without agitation, offering energy savings and operational simplicity. The findings support sustainable resource utilization and provide a practical pretreatment approach for improving biodiesel production efficiency.

Silica adsorbent, cellulose adsorbent, waste cooking oil, FFA reduction, adsorption process

EFFECT OF TEA-COFFEE WASTE RATIOS ON THE CALORIFIC VALUE OF TORREFIED BIOBRIQUETTES

Aldillah Herlambang1, Dwinka Aldian Nugraha2, Boy Setiawan3, Melani Putri4, Ayu Nadila Safitri Putri5, and Ilham Dwi Arirohman6

1 Universitas Jambi, Jambi, Indonesia
2 Institut Teknologi Sumatera, Lampung, Indonesia

Biomass residues such as spent tea leaves and coffee grounds offer significant potential as renewable solid fuels, yet their energy performance varies widely depending on composition and pre-treatment. This study investigates how different mixing ratios of tea and coffee waste influence the calorific value of biobriquettes produced through torrefaction. Five formulations (1:0, 1:1, 2:1, 1:2, and 0:1) were prepared, dried, milled, and subjected to torrefaction at 250 °C. Proximate analysis was performed to determine moisture, volatile matter, ash content, and fixed carbon, while calorific values were quantified using a bomb calorimeter. Torrefaction was found to substantially enhance fuel quality by lowering moisture and increasing fixed carbon. Coffee grounds exhibited superior fuel characteristics, yielding the highest calorific value of 5704.27 cal/g, whereas the 1:2 tea–coffee blend produced a comparable value of 5687.27 cal/g. These results indicate that blended compositions can achieve high energy performance while improving resource flexibility. Overall, the study demonstrates that torrefied mixtures of tea and coffee waste represent a promising pathway for developing sustainable solid fuels within waste-to-energy applications.

Biomass, Torrefaction, Calorific Value, Biobriquette, Waste-to-Energy

HORIZONTAL AXIS MULTI-BLADE WIND POWER PLANT: A CASE STUDY IN TALANG, SOLOK REGENCY

Fiqri Al Faruqi, Andre Rabiula, Dasrinal Tessal, Akhiyar Waladi, Lailal Gusri, Ali Satria Wijaya, Imas Ning Zhafarina, Aldony Reco Putra

Universitas Jambi

This study investigates the design and performance of a multi-blade horizontal axis wind turbine as an alternative energy source in Talang, Solok Regency. The main objective is to develop a wind power generator capable of operating efficiently under the low wind speed conditions typical of the Talang area, while contributing to local electricity supply. The research methodology includes six months of field wind speed measurements at multiple hub heights to establish the wind profile, analysis of regional meteorological data, prototype design with considerations of blade number, pitch angle, and rotor diameter, and evaluation of turbine performance in terms of electrical output, power coefficient, start-up (cut-in) speed, and operational stability under varying wind and load conditions. The results show that with an optimized multi-blade configuration, the turbine achieves cut in at approximately 4-5 m/s, delivering average electrical output within ±10% of the predicted values. Efficiency levels are deemed satisfactory for local deployment, with the turbine maintaining stable operation across wind speed and load variations. Innovations include the adaptation of blade geometry suitable for low-speed wind, use of cost-effective materials, and design for low maintenance. This case study demonstrates that multi-blade horizontal axis wind turbines are viable for regions like Talang with modest wind resources. Recommendations highlight further refinement of blade design and integration of control systems for enhanced consistency and scalability in rural or remote settings.

Multi-Blade Wind Turbine, Horizontal Axis, Low Wind Speed, Cut-in Speed, Talang Solok, Renewable Energy

CHARACTERIZATION OF DAILY SOLAR RADIATION INTENSITY BASED ON IN-SITU DATA USING THE BH1750 SENSOR

Jesi Pebralia, Yoza Fendriani, M. Ficky Afrianto, Frastica Deswardani, Alrizal

Universitas Jambi

This study aims to characterize the daily solar radiation intensity using in-situ data obtained from the BH1750 sensor. The measurement system was designed using the BH1750 light sensor, with data transmitted wirelessly to the ThingSpeak cloud platform for storage and analysis. Data collection was carried out over two consecutive days in an open area with a recording interval of one hour. The measurement results show that the intensity pattern on the first day reached its peak (approximately 62 kLx) at 12:00 noon. This pattern exhibits a nearly symmetric distribution, representing relatively clear and stable weather conditions around midday. In contrast, the second day's peak intensity (around 65 kLx) occurred earlier, at 11:00, and rose again at 13:00, indicating sharp intensity fluctuations. These fluctuations resulted in two distinct local peaks before the intensity declined to zero at sunset. The differences in peak timing and daily fluctuation patterns highlight the need for further analysis using machine learning approaches to accurately model the complex and non-linear relationships within solar radiation data.

Radiation Intensity; Sensor BH1750; Machine Learning

APPLICATION OF NI-CARBON CATALYST IN THE CONVERSION OF USED COOKING OIL FROM INSTANT NOODLE FRYING PROCESSES INTO BIOFUEL

Nazarudin (1,2), Ulyarti (2,3), Mursalin (2,3), Hadistya Suryadri (1,2)

1 Chemical Engineering Department, Faculty of Sciences and Engineering, University of Jambi,Indonesia
2 Centre of Excellence on Bio-Geo Material and Energy, University of Jambi, Indonesia
3. Acricultural Technology Department, Faculty of Agriculture, University of Jambi,Indonesia

A review of previous research studies (1-21) shows that Nickel metal as a catalyst performs well for cracking reactions and can prevent coke formation. Previous studies have shown that the method of developing Nickel onto carbon by impregnation and ion-exchange is known to have a small amount of Ni entering the carrier (5, 8, 9, 11, 15, 20, 21). The use of the hydrothermal method shows a significant increase in the amount of Nickel carried in carbon, with the increase in the amount of Nickel in carbon resulting in an increase in the conversion of used cooking oil into biofuel (22). However, a review of previous studies shows that used cooking oil sourced from households/restaurants contains various types of chemicals originating from the extraction of fried food ingredients so that when cracked it produces biofuel containing several undesirable chemical compounds such as acetic acid and carboxylic acid (7, 14, 22, 27). In this study, the results of previous innovations on the synthesis of Ni-Carbon pellet catalysts will be developed in the cracking of used cooking oil using used cooking oil samples/feeds that only come from used frying instant noodles, to ensure that chemical impurities are only extracted from the ingredients for making instant noodles. Another reason, based on data from the Central Statistics Agency in 2024, the average consumption of instant noodles in Indonesia per capita/week is 1 pack (80 grams or total production per year of 20 billion packs (28) which of course produces used cooking oil that has the potential to be processed into biofuel. In this study, the synthesis and characterization of Ni-carbon Pellet catalysts will be carried out using methods that have been discovered previously, making instant noodles where the remaining cooking oil is used as a sample of used cooking oil which is then cracked using a Ni-carbon Pellet catalyst to become biofuel.

Instant Noodle, Cracking, Ni-Activated Carbon, Pellet,Used Cooking Oil

IOT-BASED LOW-COST SENSORS DEVELOPMENT AS AN ALTERNATIVE FOR CITY'S AMBIENT PM2.5 MONITORING IN INDONESIA

Rizki Andre Handika1, Yosi Riduas Hais1, Damris Muhammad1, Febri Juita Anggraini1, Kemas Rahmat Wiharja2, Iqbal Mutawaqil1, Putri Ayu Risti1, Aida Khairani1, Rahmi Yulia Fitri1, and Tegar Rizki Saputra1*

1 Faculty of science & Technology, Universitas Jambi, Jambi 36361, Indonesia
2 School of Computing, Telkom University, Bandung 40257, Indonesia

Air quality has become a critical environmental problem globally, in which ambient PM2.5 got the most consideration. Since more than 90% of the population were alive in PM2.5 concentrations above the WHO standard, efforts in air pollution monitoring and control have been recognized. The monitoring aspect emerged to be important for mitigation to attain appropriate control strategies based on the problems. However, about 1 billion rupiah is needed for one air quality monitoring station (AQMS), making it still unavailable for each city in Indonesia. Moreover, numerous AQMS is needed to have comprehensive monitoring in Indonesian cities. This paper presents recent progress of an Internet of Things (IoT)-based low-cost sensor (LCS) development that is still studied in Jambi city as an alternative AQMS solution for ambient PM2.5 monitoring. Our two-year work and achievements will be reported, which were started from aurduino-based to ESP32 microcontrollers with some trial LCSs (from GP2Y1010AU, ZH03B, PMS5003, until PMS7003 sensors), while the next work planned will be discussed with focusing on accuration and validation. The proposed system harnesses affordable optical sensors with IoT connectivity, emphasizing robust calibration, real-time data delivery, and cloud-based visualization adaptable to local atmospheric conditions.

PM2.5 pollutant, AQMS, low-cost sensor, IoT-based, city levels.

REVITALIZING COMMUNITY LIVELIHOOD THROUGH EMPOWERMENT AND MANAGEMENT OF LUBUK LARANGAN GUCI EMAS

Tedjo Sukmono
Sigit Indrawijaya
Lauura Hermala Yunita
Yoppie Wulanda
Rizky Janatul Magwa
Febrina Rolin
Yusyam Leni
Tri Syukria Putra
Pradita Eko Prasetyo Utomo

Universitas Jambi

The Batanghari River has experienced a decline in fish diversity from around 300 to 100 species between 1990 and 2022, indicating ecological degradation. Lubuk Larangan Guci Emas in Muaro Pijoan was designated in 2022 as a local wisdom-based conservation and tourism area. However, management, facilities, and promotion remain major challenges. This community service program aims to enhance the capacity of the Tourism Awareness Group (POKDARWIS) in managing fish conservation and community-based ecotourism through training in management, marketing, and the application of innovative technologies such as the Ber-Ikan Jambi application, fiber boats, and life jackets. The results showed a significant increase in management and marketing scores from 10.2 to 17.1, as well as improved income and community independence. The program contributes to institutional strengthening, technological application for conservation, and sustainable nature-based tourism development.

Lubuk Larangan Guci Emas Community Empowerment Fish Conservation Nature-Based Tourism

IMPLEMENTATION OF HOUSEHOLD ORGANIC WASTE PROCESSOR USING AEROBIC AND ANAEROBIC FERMENTATION METHODS

Bagus Adhitya, Muhammad Ikrar Lagowa, Aditya Denny Prabawa, Jarot Wiratama, Yosa Megasukma, and Maulina Tanjung

Teknik Kebumian, Fakultas Sains dan Teknologi, Universitas Jambi

The problem of household organic waste management has become a serious challenge in many regions of Indonesia, including RT 10 of Sungai Duren Village, Muaro Jambi Regency. Most residents still rely on traditional methods such as burning or disposing of waste in vacant land, which leads to environmental pollution, foul odors, and serves as a breeding ground for disease vectors such as flies and rats. In fact, more than 57% of the total 67.8 million tons of national waste per year consists of organic waste that can actually be processed into useful products. The main obstacles faced by the community are the lack of appropriate waste-processing equipment designs and low literacy in waste management. This community service program aims to design and implement a household-scale organic waste processing device that integrates aerobic and anaerobic fermentation in a single, unified system. The implementation method uses Focus Group Discussion (FGD) with a participatory approach so that the community can be actively involved in all stages of the activity, which include: preparation and initial survey, design and fabrication process, socialization, as well as monitoring and evaluation. Monitoring is carried out using an online survey via Google Forms to assess the effectiveness, ease of use, and sustainability of the device. The results show that the device functions effectively in reducing the volume of household organic waste and produces compost and biogas that can be reused. In addition, there is an increase in community knowledge, awareness, and participation in independent, sustainable, and environmentally friendly waste management.

Organic_waste; Composter; Aerobic–anaerobic_fermentation

DEVELOPMENT OF AN ANDROID-BASED EDUCATIONAL GAME FOR THE PRESERVATION OF JAMBI FOLKLORE: A CASE STUDY OF THE LEGEND OF THE 'ORANG KAYO HITAM'

Dedy Setiawan, Tri Suratno, Muhammad Razi A., Dewi Lestari

Universitas Jambi

This research aligns with the University of Jambi's Strategic Plan (Renstra UNJA) for 2024–2028, which emphasizes the use of digital technology and the preservation of local wisdom through research-based innovation.

One implementation is the development of educational media that can introduce local cultural values in an engaging and contextual manner. In this context, the development of an Android-based educational game based on the folktale "Orang Kayo Hitam" is a response to the increasing need for effective digital learning media to instill cultural values and moral character in children. With the advancement of digital technology, game-based media has proven to be an innovative solution for increasing children's interest in learning and understanding of culturally based materials. This research aims to design and develop an educational game application using Unity as a development platform, using a Multimedia Development Life Cycle (MDLC) software engineering approach. In addition to design, this study also evaluates the application's quality based on four software quality standards: functionality suitability, performance efficiency, compatibility, and usability. Data collection was conducted through a literature review to obtain a theoretical foundation and a questionnaire was distributed to users after the application was completed. The analysis involved students from the Ash-Shiddiiqi IT Elementary School in Jambi City to test the application's functionality and performance efficiency. Quantitative evaluation of usability aspects was presented as a percentage as the basis for assessing the application's feasibility. The outputs of this research include:
(1) the creation of interactive media based on local culture; (2) a scientific article for publication in the Business Information Systems Journal (JSINBIS) - https://ejournal.undip.ac.id/index.php/jsinbis or other SINTA 2 journals; and (3) a presentation of the research results at the 2025 JICEST Seminar at the Faculty of Science and Technology, University of Jambi.

Game; Android; MDLC; Folklore; Jambi

ANALYSIS OF $PM_{2.5}$ POLLUTANT BACKWARD TRAJECTORIES IN JAMBI CITY USING THE HYSPLIT MODEL

1. Nesly Oktaviani Saragi Napitu
2. Febri Juita Anggraini, S.T., M.T.
3. Zuli Rodhiyah, S.Si,. M.T.

Jambi University

Air pollution in Jambi City shows fluctuations in air quality due to various sources of pollution, both from local activities and long-distance pollutant transportation. In 2024, the average concentration of $PM_{2.5}$ was recorded to have exceeded the annual quality standard of 19 $\mu g/m^3$, an increase compared to the previous two years, which reached 16.11 $\mu g/m^3$ and 8.37 $\mu g/m^3$, respectively. This increase occurred after the COVID-19 pandemic due to changes in people's activities returning to normal. Using the HYSPLIT Backward Trajectory Model, this study identified the potential areas of origin of $PM_{2.5}$ pollutant backward trajectories in Jambi City before, during, and after the pandemic, covering the years 2018 to 2024. The results of the study show that before and after the pandemic, the dominant pollutants originated from Ogan Komering Ilir Regency and Banyuasin Regency, especially in months with low rainfall, such as July. Meanwhile, during the pandemic, the potential areas of origin of the backward trajectory of pollutants were more varied, covering Malaysian waters, Muara Enim Regency, Tanjung Jabung Barat Regency, Bintan Regency, and several other areas.

$PM_{2.5}$, air quality, HYSPLIT, COVID-19, Jambi City

ETHNOSCIENCE OF ENAU (ARENGA PINNATA): EXPLORING THE LOCAL KNOWLEDGE OF JAMBI COMMUNITIES FOR CONTEXTUAL SCIENCE LEARNING

Bambang Hariyadi, Azizah Turrohmah, Winda D. Kartika, Cut Multahadah, Rahma Dani, Danial Mursyid, Fitri Astriawati, Ine Tentia

Biology Program, Faculty of Science and Technology, Universitas Jambi
Biology Education Program, Faculty of Education and Techer Training, Universitas Jambi
Mathematics Program, Faculty of Science and Technology, Universitas Jambi
Physics Education Program, Faculty of Education and Techer Training, Universitas Jambi

Several traditional communities possess rich local knowledge and traditions related to the management of the sugar palm (Arenga pinnata), encompassing ecological, economic, and cultural values. However, much of this knowledge remains undocumented and has not been optimally utilized in science education. This study aims to explore and document local community knowledge about the sugar palm and to examine the ethnoscientific values embedded within it for potential integration into contextual biology learning. The research was conducted in Jambi Tulo Village, Muaro Jambi Regency, using a descriptive qualitative approach. Informants were selected purposively and expanded using snowball sampling, involving sap tappers, palm sugar producers, customary leaders, and community members possessing traditional knowledge related to the sugar palm. Data were collected through participant observation, in-depth interviews, visual documentation, and literature review, and analyzed using thematic and content analysis to identify relevant scientific concepts. The results indicate that the traditional management of the sugar palm reflects an integration between local knowledge and biological concepts such as plant structure and function, physiological processes (osmosis, root pressure), as well as ecological and conservation principles. Integrating these ethnoscientific values into biology learning can enrich learning contexts, strengthen students' scientific understanding, and foster ecological awareness and appreciation of local wisdom.

Arenga pinnata; local knowledge; contextual learning; ethnobotany; ethnoscience

EMPOWERING SCIENCE PRACTICUM THROUGH SMARTPHONE-BASED DIGITAL IMAGE ANALYSIS: A CONTEXTUAL LEARNING APPROACH

Martina Asti Rahayu, Arum Widia Putri, Dina Erliana, Indra Lasmana Tarigan, Dhian Eka Wijaya, Ayu Nadhila Safitri

Universitas Jambi

Science practicum plays a vital role in strengthening students' conceptual understanding; however, limitations in laboratory facilities often hinder its optimal implementation. To address this challenge, this community service program was designed to enhance practical science learning through digital innovation using smartphones and ImageJ software. The program was conducted at SMA Negeri 13 Jambi as an example of contextual and technology-based practicum enhancement.

The main objective of this program was to empower teachers and students to utilize smartphone cameras and digital image analysis for the detection of water pollution, serving as a real-world application of analytical chemistry concepts. The implementation included workshops, demonstrations, and guided practice sessions on capturing water sample images using smartphones and processing them with ImageJ. Through this approach, students learned to analyze RGB intensity variations to determine differences in water turbidity and color, which represent the presence of pollutants. The results indicated a significant improvement in students' engagement, scientific reasoning, and digital literacy. Teachers also reported that the method provided a feasible and low-cost alternative to conventional laboratory instruments while maintaining analytical relevance. In conclusion, this program successfully strengthened science practicum implementation through the integration of smartphone-based digital tools and contextual environmental analysis, promoting a more interactive, accessible, and technology-oriented learning experience in secondary education.

contextual practicum, smartphone-based innovation, ImageJ, digital image analysis, science education

PORTABLE GRAM-STAIN KIT: ENABLING INQUIRY-BASED MICROBIOLOGY IN HIGH SCHOOLS

Fitra Wahyuni1, Bambang Hariyadi2, Tedjo Sukmono3, Mahya Ihsan4, Anggit Prima Nugraha5, Uni Baroroh Husnudin6, Andini Vermita Bestari7 and Ashif Irvan Yusuf8

Universitas Jambi

Many Indonesian senior high schools lack functional biology laboratories, limiting experiential microbiology learning. We developed a portable Gram-stain kit plus guided training to enable hands-on practicals aligned with inquiry-based curricula. Using a five-stage community-engagement model (co-planning; kit prototyping/validation; teacher–student workshops; monitoring/evaluation; dissemination), we delivered 30 ready-to-use kits, and a concise step-by-step module to SMAN 3 Muaro Jambi. Students (small groups) completed the full Gram-stain workflow (smear, heat fixation, crystal violet, iodine, alcohol decolorization, safranin), followed by microscopy and interpretation. Learning outcomes were assessed with paired pre/post multiple-choice tests (10 items), procedural checklists, and user feedback. Paired analyses used Shapiro–Wilk for normality and Wilcoxon signed-rank when appropriate ($\alpha=0.05$). All groups produced interpretable slides (purple Gram-positive; pink Gram-negative), demonstrating feasibility and face validity of the kit. Aggregate knowledge gains were moderate (post-test > pre-test by ~6–10 points on a 0–100 scale) but not statistically significant (Wilcoxon, $p>0.05$), constrained by ceiling effects (high baseline scores). Qualitative data showed increased confidence, strong engagement, and teacher readiness to replicate the practical independently using remaining kits. The portable Gram-stain kit is a fit-for-purpose, scalable, and low-cost educational technology that enables authentic microbiology experiences in resource-constrained schools. While aggregate cognitive gains were limited by high baselines, procedural and affective outcomes and technology transfer to teachers were substantial. Future work will extend to additional low-cost microbiology kits, strengthen longitudinal monitoring, and scale dissemination via district partnerships.

Gram stain; portable lab kit; inquiry-based learning; experiential science practicals; resource-constrained high schools (Indonesia).

FROM LOCAL ADAPTATION TO NATURE-BASED SOLUTIONS IN THE SOCIO-ECOLOGICAL TRANSFORMATION OF SAYUNG, DEMAK

Retno Sari Dewi
Elma Mutmainnah
Akbar Wira
Ayu Aini Rachmawati
M. Shulhan Jihadi

Department of Environmental Engineering Universitas Diponegoro, Semarang, Indonesia
Center for Urban and Regional Resilience Research Universitas Diponegoro, Semarang, Indonesia
Department of Urban and Regional Planning Universitas Diponegoro, Semarang, Indonesia
Department of Urban and Regional Planning Universitas Diponegoro, Semarang, Indonesia
Department of Marine Science Universitas Diponegoro, Semarang, Indonesia

The northern coastal region of Java Island has undergone rapid socio-ecological transformations due to tidal flooding, coastal abrasion, and land subsidence. This study examines the dynamics of the coastal ecotone in Sayung, Demak Regency, highlighting the shift from local adaptive strategies toward the implementation of Nature-Based Solutions (NbS). Field observations and direct interviews were conducted in two villages—Wonorejo and Timbulsloko—on September 20, 2025. The results indicate significant landscape changes, starting from rice fields that were converted into aquaculture ponds and eventually transformed into mangrove conservation areas. Local communities have developed various adaptation strategies, such as building floating settlements, elevating houses, and conducting mangrove rehabilitation. However, persistent environmental pressures have gradually exceeded the communities' adaptive capacity, leading to social and economic vulnerabilities. The findings emphasize the importance of integrating local knowledge with the NbS concept to enhance resilience and restore ecological functions in degraded coastal areas. Furthermore, this study contributes to understanding adaptive transformation processes in Indonesian coastal ecotones and supports the development of community-based coastal management frameworks.

Coastal Ecotone; Sayung; Community Adaptation; Land Subsidence; Nature-Based Solutions

EVALUATION OF LEAK ELEVATION EFFECTS ON AMMONIA STORAGE RELEASE CONSEQUENCES USING ALOHA SIMULATION

Aldillah Herlambang, Beatric Erlina Sitohang

University of Jambi, Jambi, Indonesia

communities due to its toxic and flammable properties. This study investigates the influence of leak hole elevation on Quantitative Risk Assessment (QRA) outcomes for the 5,000 MT Ammonia Storage Tank 4101-F at PT Pupuk Sriwidjaja Palembang, located near residential areas. Simulations were conducted using ALOHA 5.4.7 software to compare two scenarios: low elevation leak (0.5 m above ground) and high elevation leak (10 m above ground), with a 3-inch diameter hole under worst-case meteorological conditions (stability class F, wind speed 2 m/s). Four consequence scenarios were evaluated: toxic vapor cloud, flammable vapor cloud, vapor cloud explosion, and BLEVE. Results demonstrated that low elevation leaks generated significantly higher release rates (1,820 kg/min) compared to high elevation leaks (974 kg/min) due to greater hydrostatic pressure. The toxic vapor cloud ERPG-3 zone extended to 1.4 km for low elevation versus 0.953 km for high elevation. Similarly, flammable vapor cloud LEL zones reached 156 m and 105 m, while vapor cloud explosion overpressure (1 psi) zones extended to 168 m and 114 m, respectively. The findings reveal that leak elevation substantially affects consequence dispersion patterns and hazard zone extents, providing critical insights for emergency response planning and safety system optimization in ammonia storage facilities.

Ammonia Storage; Quantitative Risk Assessment; Leak Elevation; ALOHA Simulation; Consequence Analysis

IDENTIFICATION OF ACTIVE FAULT TRACES OF "PALEO EARTHQUAKES" USING GRAVITY METHODS IN JANGKAT, MERANGIN REGENCY

Anggi Deliana Siregar
D. M. Magdalena Ritonga
Sarwo Sucitra Amin
Maulina Tanjung
Shazkya Annura Rizky
Pius Lauren Datu Bara

Universitas Jambi

Indonesia is a country with high seismic activity because it is located at the meeting point of four major tectonic plates, resulting in numerous active faults and susceptibility to geological disasters such as earthquakes and landslides. Despite the high risk, data on active faults and geological disasters in Indonesia is still limited. This study focuses on geological mapping to identify Active Fault Traces, specifically "Paleo Earthquakes," in Jangkat District, Jambi Province. The purpose of this study is to update and verify the position of faults from old geological maps, as well as to interpret new deformation mechanisms resulting from the interaction of geological structures in the field. Through preliminary studies and analysis of field data, as well as secondary geophysical data. Based on the results of geological structure measurements in the field, there are three fault locations consisting of a left-lateral strike-slip fault, a right-lateral strike-slip fault, and a right-lateral strike-slip fault. Analysis of the residual map shows residual gravity anomalies with amplitudes of -9.6 to +11.0 mGal. The dominance of positive anomalies in the southwest is interpreted as an active fault and is significantly consistent with the distribution of active faults identified in the field. A linear residual anomaly pattern with a relatively high northwest-southeast gradient that correlates directly with the fault segment can be seen on the alignment map, which shows the general direction pattern. The results of the research location analysis are traces of active faults from paleo-earthquakes that have been identified through field and gravity data.

Active Faults, Geological Disasters, Gravity

TECHNOLOGICAL EVALUATION OF THE IMPLEMENTATION OF ADVANCED PROCESS CONTROL AT THE RFCC UNIT OF PT KILANG PERTAMINA INTERNASIONAL

Suhar Prianto1, Andri Wiyo2, and Teguh Ariyanto3*

1Engineer professional Program Study Program, Faculty of Engineering, Gadjah Mada University, Yogyakarta
2Engineering & Development Department, PT. KPI RU IV, Cilacap
3Department of Chemical Engineering, Faculty of Engineering, Gadjah Mada University, Yogyakarta

The RFCC (Resid Fluid Catalytic Cracking) unit has a design capacity of 62 MBSD, with raw oil feed coming from Low Sulphur Wax Residue (LSWR) ex CDU II and Vacuum Gas Oil ex-HVU I/II. Raw oil feed will go through the cracking process using a fluidized catalyst into a lighter and more valuable product, including: Propylene products are produced in unit 104, Propylene Recovery as an overhead product C3 Splitter, after going through the advanced treatment process of moisture removal in the propylene drier and COS and metal removal in the propylene treater. RFCC naphtha derived from unit 101/102 removed sulphur contaminants and ditreated external diolefin fraction in unit 105 Gasoline Hydrotreating (Prime-G) produces desulphurized RFCC gasoline. Ex-RFCC gasoline products are high octane blending (HOMC) components, flowed to the RFCC gasoline tank. LPG (mixed C3/C4) products are produced in unit 104, Propylene Recovery after going through the process of treating and separating in unit 102, Gas Concentration and unit 103, LPG Merox. In unit 104, LPG is produced as the bottom product of C3/C4 Splitter and C3 Splitter. Light Cycle Oil (LCO) received from the LCO Product Coalescer at RFCC Unit 101 will be delivered to the existing MFO header or LCO Tank and can also be delivered to the Fuel Oil Tank as a blending component of the refinery fuel oil (RFO). Decant Oil (DCO) is the bottom distillate of RFCC products. Before going to the product tank, the fine catalyst is first separated using an automatic backwash filter. Some DCO can also be sent to the Fuel Oil Tank as a blending component of the refinery fuel oil (RFO). Heavy Cycle Oil (HCO) is an intermediate product that is used to circulate back to the reactor to be used for heat, besides that HCO can also be reused as reactor feed, under certain conditions. The problems that occur in RFCC operations are as follows: Fluctuations in reactor outlet temperature which has an impact on fluctuations in RFCC conversion, Fluctuations in temperature profiles in the main column which have an impact on suboptimal valuable products produced., Tuning operating conditions that take a long time in Unit 104 to obtain high purity propane in conditions while maintaining on-spec propylene quality. There is still room for improvement in the RFCC unit to get more optimal operating conditions to improve the stability of operating conditions at RFCC PT KPI RU IV Cilacap so that it is able to produce products to the maximum, one of the optimization efforts that can be done is to implement Advanced Process Control (APC) in the RFCC unit, whose use has been widely applied to increase profits through the optimization of valuable products, efficiency, and energy saving.

APC, Advanced Process Control, optimization

REVITALIZING FACULTY OPEN SPACE THROUGH LOCAL FLORA AND TECHNOLOGICAL INTEGRATION

Mahya Ihsan, Bambang Hariyadi, Fetty Febriasti Bahar, Nurza Purwa Abiyoga, Andre Rabiula

Universitas Jambi

Green open spaces within university environments play a crucial role in enhancing academic, social, and ecological quality. At the Faculty of Science and Technology, Universitas Jambi (FST UNJA), the existing park has not yet functioned optimally due to a non-functional landscape layout, unsystematic use of local flora, and the absence of technological integration to support sustainability. These conditions have resulted in limited social interaction, inadequate environmental education functions, and a lack of intelligent systems for efficient green space management. Therefore, revitalizing the faculty's open space has become an urgent initiative to realize the university's vision of creating a green, innovative, and competitive academic environment. This study aims to design, implement, and integrate a technology-based green park (smart eco-park) that functions as a communal area, an interdisciplinary learning platform, and a pilot model for sustainable open-space management. Using an applied research design combining qualitative and quantitative methods, the study comprises four stages: pre-design analysis and surveys, landscape and technological system design, physical implementation, and subsequent evaluation. The main innovations include the use of local flora, application of smart irrigation systems, humidity sensors, and plant QR codes to enhance user engagement and data-driven environmental learning. This research contributes to scientific advancement and physics education by demonstrating the practical application of sensor-based monitoring systems, environmental data acquisition, and energy-efficient technologies to support sustainability and interdisciplinary learning within a university setting.

Campus Green Open Space; Landscape Revitalization; Local Flora; Smart Eco-Park; Smart Technology

AUTO SWITCHING WIRELESS NETWORK ALGORITHM FOR SINGLE-INTERFACE ARMBIAN DEVICES

Khairul Alim, Niken Rarasati, Yuliana Safitri, Muhammad Razi A.

Universitas Jambi

The development of information technology has increased the demand for devices capable of connecting to the internet flexibly, including in mobile environments. Single Board Computers (SBCs) such as Raspberry Pi and Orange Pi running lightweight operating systems like Armbian are widely used in embedded systems due to their stability and efficiency on ARM-based devices. However, devices with a single WiFi interface often face difficulties when switching networks automatically, which becomes a challenge in Internet of Things (IoT), robotics, and edge computing systems that require stable and continuous connections. This study aims to design an automatic WiFi network switching algorithm for Armbian devices with a single wireless interface. The proposed method involves designing an algorithm that monitors signal strength (RSSI), speed, and connection stability, which is then implemented as a bash script that runs automatically in the system background. The results show that the developed algorithm can perform network switching automatically with satisfactory transition time and in accordance with the defined parameters. Therefore, this solution enhances network reliability on Armbian devices without adding significant system overhead.

Armbian, WiFi auto switching, single board computer, bash script, network reliability

ANALYSIS OF AIR QUALITY IN JAMBI CITY BASED ON AIR POLLUTION STANDARD INDEX (ISPU)

*Tyas Hanifatunnisa, Arifin Abdullah, Bambang Irawan, and Rizki Andre Handika**

Jambi University

This study aims to analyze air quality in Jambi City using the Air Pollution Standard Index (ISPU) for PM10 and PM2,5 parameters during the 2023–2024 period. Morever, it identifies air pollution trends and their threats to public health and environmental management. A descriptive quantitative approach was employed using secondary data from the Air Quality Monitoring System (AQMS), which were subsequently converted into ISPU values in accordance with Ministry of Environment and Forestry Regulation No. 14/2020. The findings indicate that air quality in 2024 improved compared to 2023, particularly for PM_{10}, which consistently fell within the "good" category. However, PM2,5 still recorded numerous days in the "moderate" category, although it no longer reached the "unhealthy" levels observed during the peak pollution episode in 2023, which was influenced by forest and land fires. The Wilcoxon test confirmed a significant difference in ISPU values between the two years ($p < 0.001$), suggesting a measurable shift in air quality. The novelty of this study lies in its comparative analysis of two recent years, integrating ISPU conversion with inferential statistical assessment to provide a more comprehensive understanding of particulate pollution patterns in a tropical city highly vulnerable to forest and land fire (karhutla) events. These findings contribute to strengthening the scientific evidence base for pollution-control policymaking and support the development of data-driven early warning systems and mitigation strategies at the regional level.

air quality; PM10; PM2,5; ISPU; Jambi City.

GROUNDWATER MODELING USING KRIGING INTERPOLATION TECHNIQUE WITH GEOLECTRICAL TEST DATABASE

1. Muhammad Guntur, M.Si (muhammadguntur@unja.ac.id)
2. M. Farandika Akbar, M.Si (farandikaakbar@unja.ac.id)
3. Royhan Himawan Putra, S.T (royhanhimawanputra@gmail.com)

1. Program Studi Teknik Lingkungan Fakultas Sains dan Teknologi Universitas Jambi
2. Program Studi Agribisnis Fakultas Pertanian Universitas Jambi
3. Program Studi Teknik Lingkungan Fakultas Teknik Universitas Batanghari

Groundwater availability is one of the primary needs in a particular area nowadays. Nagari Sopan Jaya, where most of the community using groundwater for domestic needs and economic activities, including palm oil cultivation, which requires groundwater to support crop productivity and growth. This study aims to provide information on groundwater availability through contour conditions, depth, and groundwater flow patterns using GIS-based groundwater modeling with kriging interpolation techniques using geolectrical test data. The results of the study show that the contour of groundwater availability is at an elevation of 84 mdpl to 131 mdpl, then the depth of groundwater based on the model results varies from 6 meters to 15 meters from the ground surface with a groundwater flow pattern moving from east to west. In addition, the results of the geoelectric test show that most of the aquifers located along the measurement line are confined aquifers, with the water table being highly dependent on rainfall intensity. From the results of the groundwater modeling, it can be implied that in Nagari Sopan Jaya, with a minimum groundwater depth of 6 meters from the ground surface and a groundwater supply that is highly dependent on rainfall intensity, priority concern is needed considering the dependence on rainfall, which can affect groundwater availability up to 15 meters during the dry season. The results of this study are expected to serve as consideration for relevant parties and the local community in utilizing groundwater for domestic needs and productive activities in a sustainable way.

Groundwater Modelling, Interpolation Kriging, Geolectrical Test, GIS

FLOOD DISCHARGE SIMULATION IN KENALI KECIL RIVER BASIN USING SWAT AND SWMM

Freddy Ilfan, Hairul Mursalin, Lailal Gusri, Taufiq Ihsan

Jambi University, Andalas University

Land use changes in the Kenali Kecil Watershed, particularly in Kembar Lestari 1 Housing Complex, Jambi City, have increased surface runoff, resulting in the inability of the drainage system to accommodate peak discharge during heavy rainfall. This study aims to evaluate the capacity of the existing drainage system through hydrological and hydraulic analysis using the Soil Water Assessment Tool (SWAT) and Storm Water Management Model (SWMM). The data used include 10 years of rainfall records, climatological data, and drainage channel geometry. The analysis shows that the peak discharge of 12.683 m^3/s exceeds the capacity of the existing 3.6 m × 1.8 m drainage channel. SWMM simulation indicates flow inefficiency due to sediment accumulation and waste blockage. Recommended solutions include increasing the channel dimensions to 3.6 m × 3 m, or constructing infiltration wells and floodgates. The study concludes that redesigning the drainage system is necessary to mitigate flooding in the area.

Drainage; Flood; Soil Water Assessment Tool (SWAT); Storm Water Management Model (SWMM)

DIGITALIZATION OF TELUK VILLAGE TOURISM BASED ON VIRTUAL TOURS FOR SUSTAINABLE AGROTOURISM

Yolla Noverina, Muhammad Razi A, Dewi Lestari, Mohammad Ilhami, Reni Aryani, Hasby Kuswanto, Lina Kartika Paramita

Universitas Jambi

This study aims to develop a digital promotional media based on a 360° Virtual Tour as a means of introducing the agro-tourism potential of Teluk Village, Pemayung District, Batanghari Regency, Jambi Province. The development process uses the Multimedia Development Life Cycle (MDLC) method which consists of six stages, namely Concept, Design, Material Collection, Assembly, Testing, and Distribution. In the design stage, an interactive user interface and a 360° panoramic navigation flow are created and combined with multimedia content obtained from field documentation using a VR camera. The implementation utilizes the 3DVista Virtual Tour platform that allows users to explore panoramas, hotspot information, and location maps interactively. System testing using the Black Box Testing method shows that all main functions run well, including panoramic navigation, displaying location information, and multimedia playback without technical errors. This study produces a village website along with a 360° Virtual Tour that can be accessed online, which functions as a digital, educational, and promotional tool that empowers local communities to manage agro-tourism potential independently and sustainably. The novelty of this research lies in the integration of 360° immersive technology for rural tourism promotion, which contributes to the advancement of smart tourism and the digital transformation of rural areas.

Virtual Tour 3600; MDLC; Agrotourism; Website; Village Digitalization.

DEVELOPING A FORENSIC ENGINEERING MODEL FOR INVESTIGATION BUILDING CONSTRUCTION FAILURES: A LITERATURE REVIEW

Tatang Kukuh Wibawa1, Yusuf Latief2
1 Department of Civil and Environmental Engineering, Universitas Indonesia, Depok, Indonesia
2 Department of Civil and Environmental Engineering, Universitas Indonesia, Depok, Indonesia

Department of Civil and Environmental Engineering, Universitas Indonesia, Depok, Indonesia

This study presents a systematic literature review of Scopus-indexed publications (2015–2025) on forensic engineering investigations of building construction failures, conducted under the PRISMA 2020 protocol. From 139 identified articles, 38 were analyzed through bibliometric mapping (VOSviewer) and narrative meta-synthesis. The mapping of methods and failure types indicates that specific failures tend to be examined with particular analytical tools—for example, structural collapses are modeled numerically, while procedural or human-factor failures rely on Root Cause Analysis (RCA), Failure Tree Analysis (FTA), or expert judgment. Findings reveal that research remains predominantly technical, emphasizing structural and material failures analyzed via Finite Element Analysis, load testing, and visual inspection, while managerial, procedural, and legal aspects are underexplored. Three interrelated domains—technical, managerial, and regulative—emerged as key determinants of investigative effectiveness. The study proposes an integrated forensic engineering investigation framework encompassing six sequential phases: Pre-Scene Preparation and Coordination, Scene Examination/Investigation, Laboratory and Testing, Analytical Modelling, Causal Analysis, and Reporting and Knowledge Management. Within technical, investigative, and legal-regulatory domains, ensuring methodological rigor, procedural consistency, and legal defensibility, thereby advancing a standardized, multidisciplinary model for construction failure investigations.

Keywords: Building Construction Failures, Forensic Engineering, Failure Analysis, Integrated Investigation Framework

ANALYSIS OF WHATSAPP BUSINESS FEATURES ON THE EFFECTIVENESS OF MARKETING COMMUNICATIONS FOR CRAFT MSMES IN JAMBI CITY

Rudi Nata, Ari Andrianti, Miranty Yudhistira, Rahmad Ashar, and Renaldi Yulvianda

University of Jambi

This study analyzes the influence of WhatsApp Business features (Broadcast Message, Quick Reply, Catalog, and Labeling) on marketing communication effectiveness for Craft SMEs in Jambi City. Using a mixed-methods approach, quantitative data were collected through structured questionnaires from 385 creative SME respondents, while qualitative data were gathered through in-depth interviews, case studies, and focus group discussions. Multiple linear regression analysis revealed that WhatsApp Business features simultaneously significantly influence marketing communication effectiveness (R^2=0.596, p=0.003), explaining 59.6% of the variance. However, partial analysis showed only the Catalog feature had significant positive influence (β=0.903, p=0.034), while Broadcast Message and Labeling showed positive but insignificant effects, and Quick Reply demonstrated negative influence. The novelty lies in focusing specifically on creative SMEs in Jambi with cultural heritage products and comprehensively analyzing four WhatsApp Business features simultaneously. This research contributes to digital marketing literature by demonstrating how digital tools can preserve local cultural values while enhancing business competitiveness. The findings imply that creative SMEs should prioritize optimizing Catalog features with high-quality visual content that showcases cultural values, supported by government digital literacy programs tailored to heritage-based businesses.

Features, WhatsApp, Effectiveness, Communication, MSMEs

EMPOWERING LOCAL ENTREPRENEURS THROUGH DIGITAL MARKETING AND PINEAPPLE PRODUCT INNOVATION IN TANGKIT BARU VILLAGE TO SUPPORT THE NATIONAL "PROUDLY MADE IN INDONESIA" MOVEMENT

Muhammad Razi A., Ulfa Khaira, Willy Bima Alfajri, Zainil Abidin, Yosi Riduas Hais, Mutia Fadhila Putri, Dedy Setiawan, Reni Aryani

Universitas Jambi

This community service program was carried out in Tangkit Baru Village, Sungai Gelam District, Muaro Jambi Regency, which is known as one of the largest pineapple production centers in Jambi Province. The main problems faced by the community are the lack of innovation in pineapple derivative products and limited ability to market products digitally. The purpose of this activity is to enhance the capacity of local entrepreneurs through training on pineapple-based product innovation and the introduction of e-commerce platforms to expand market reach. The implementation method used participatory approaches, training, mentoring, and practical sessions. The results of the activity showed an increase in the knowledge and skills of entrepreneurs in product packaging, branding, and the use of digital media for product promotion. Additionally, new business groups were formed that utilize digital technology as a marketing tool. Therefore, this program contributes to strengthening the local economy and supports the national movement of 'Proudly Made in Indonesia.'

Empowerment, Digitalization, E-Commerce, Product Innovation, Pineapple